OPERATING SYSTEMS

(UE25CS242A)

LEVEL 3 MINI-PROJECT

CampusCompute: Intelligent Job Scheduling and Resource Management

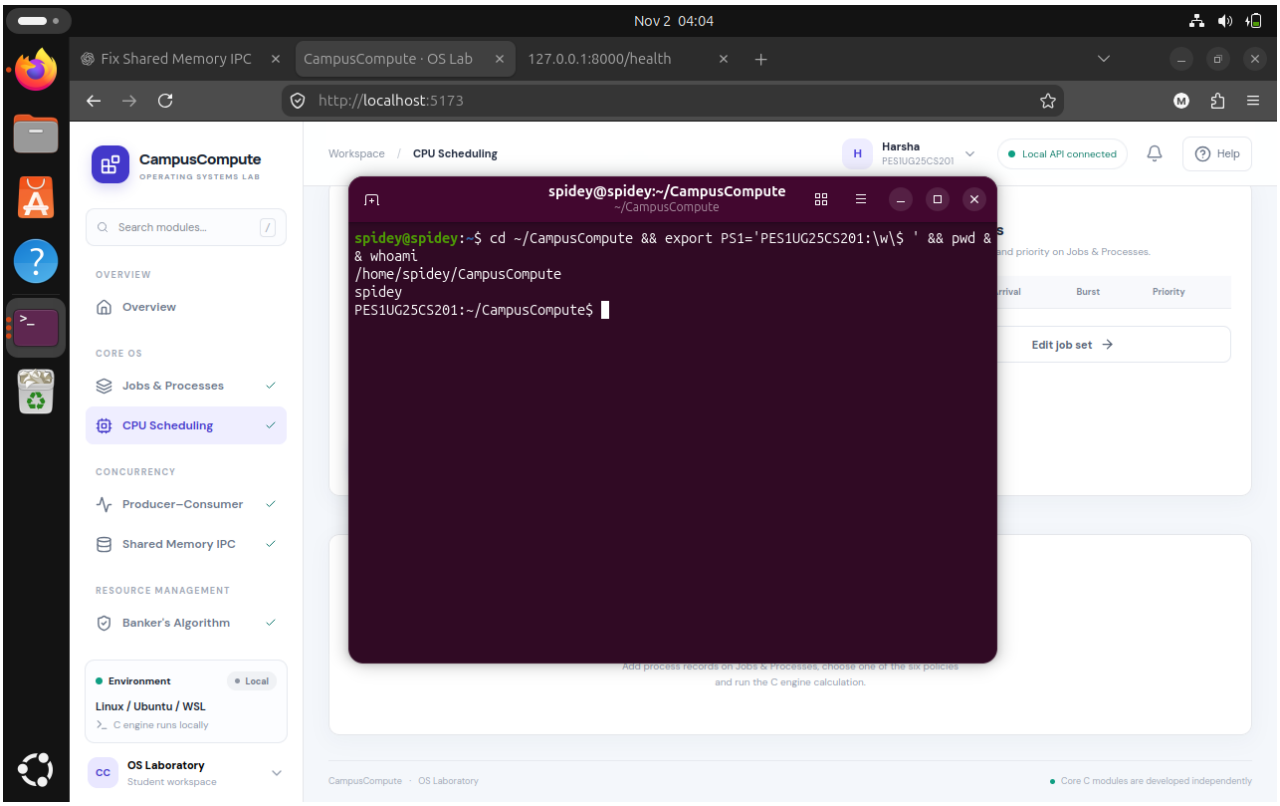
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Screenshot Evidence (S01 – S24)

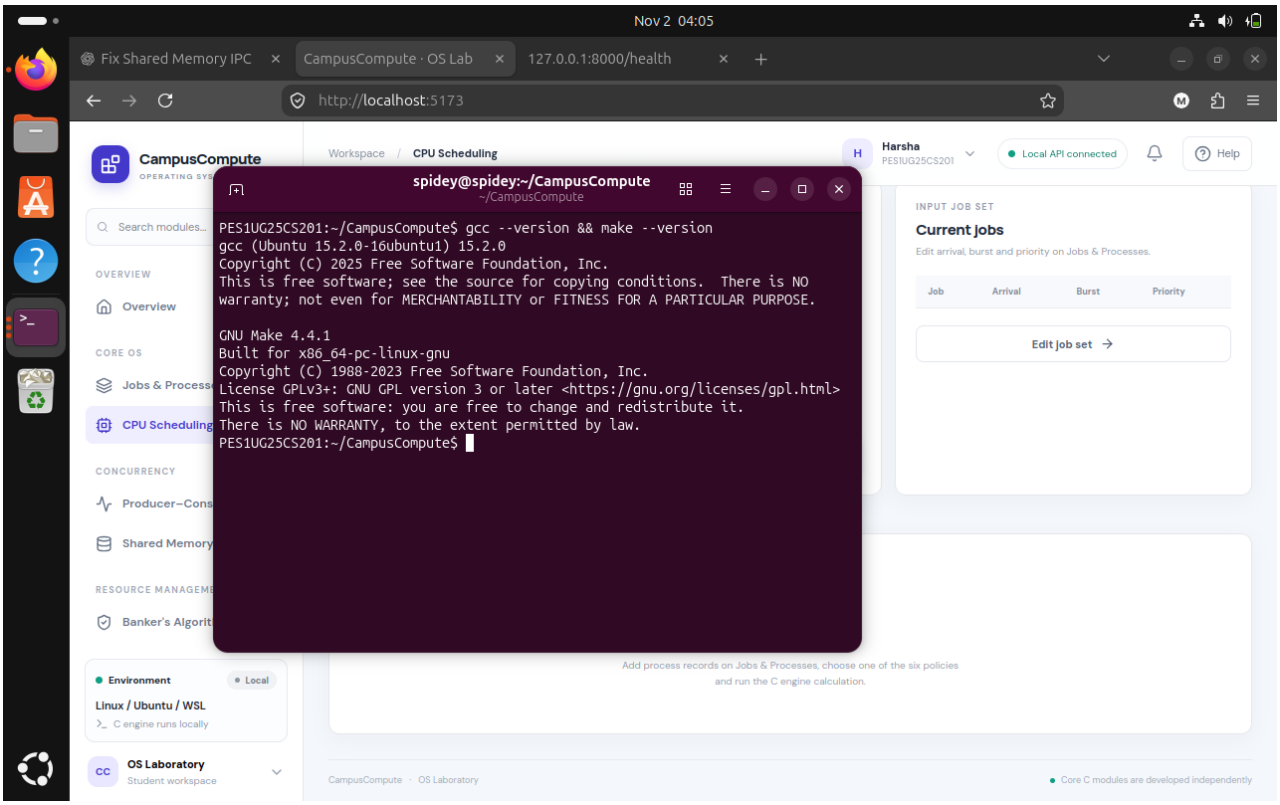
Team Members

Member	Name	SRN
Member 1	Harshavardhan M	PES1UG25CS201
Member 2	Nischal.k	PES1UG26CS834

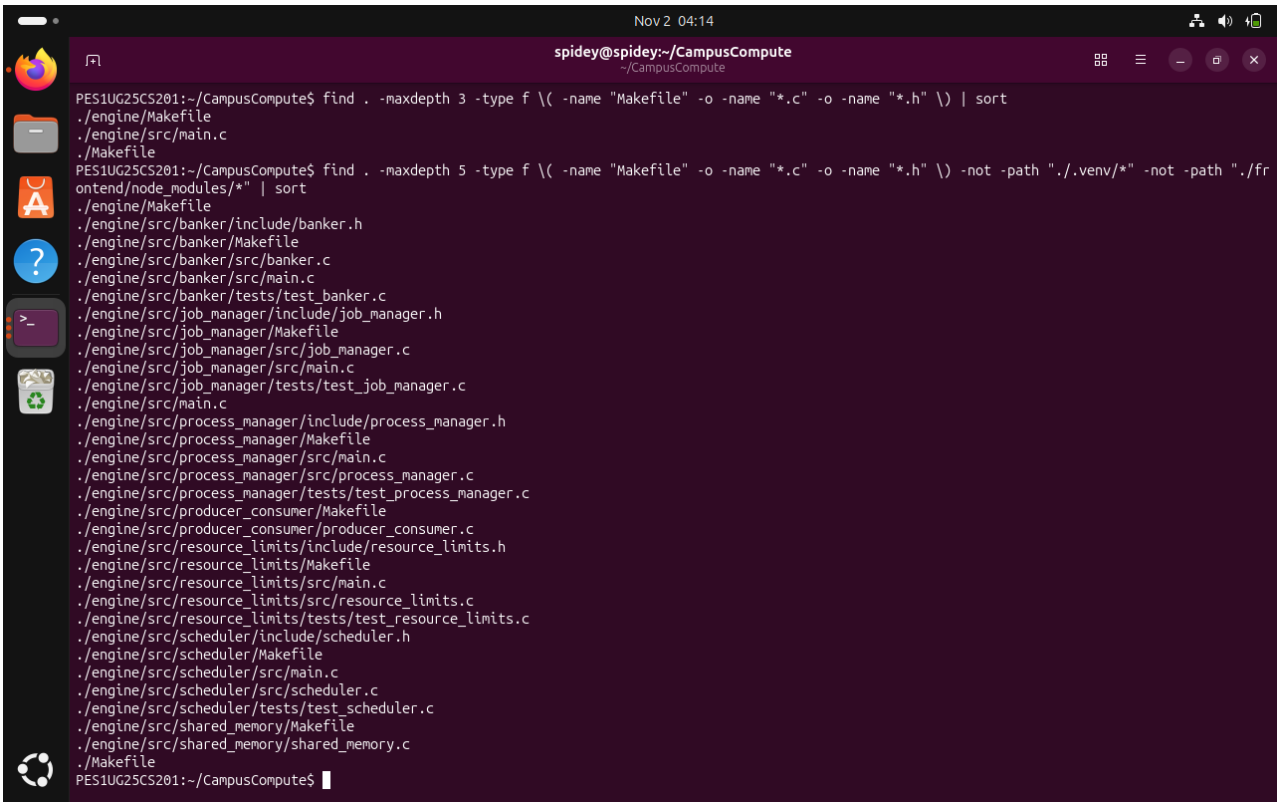
S01: Ubuntu terminal showing the full SRN prompt, the user and the working directory ~/CampusCompute.



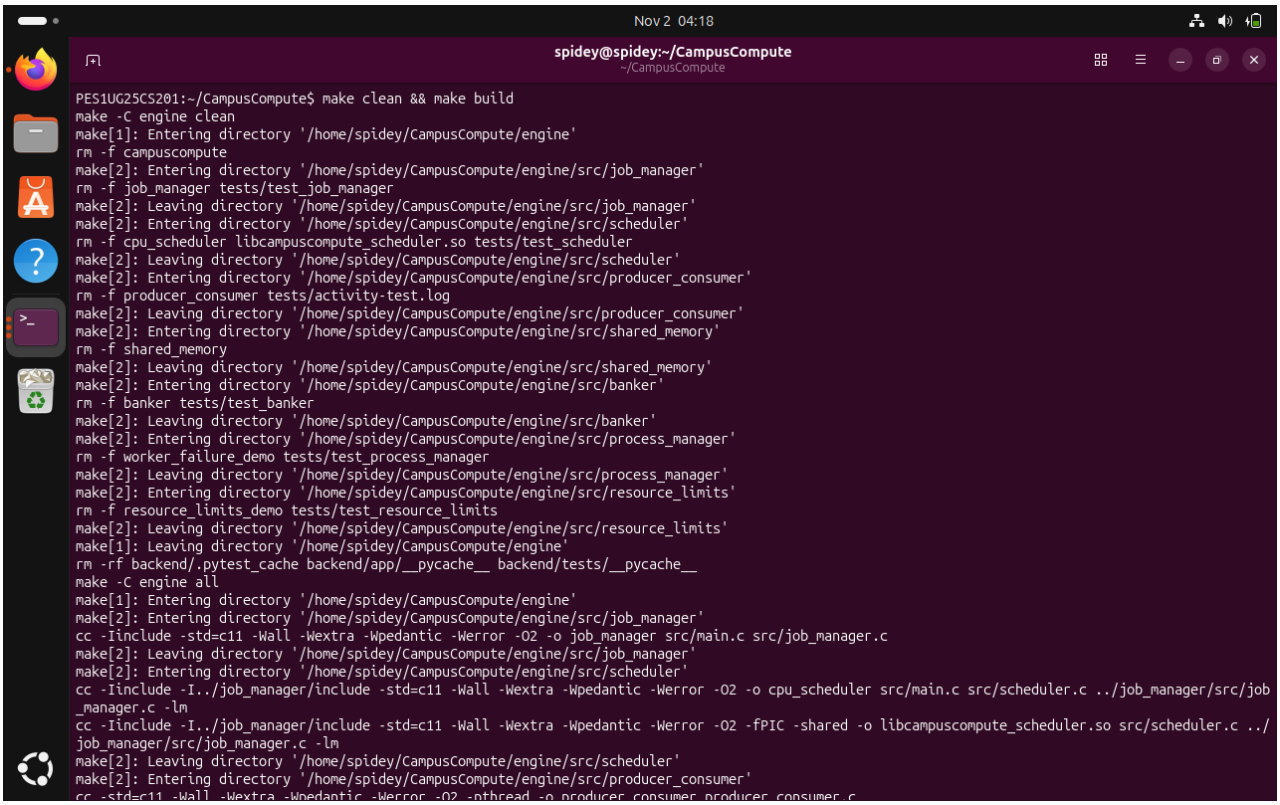
S02: GCC 15.2.0 and GNU Make 4.4.1 versions confirm that the required build tools are available.



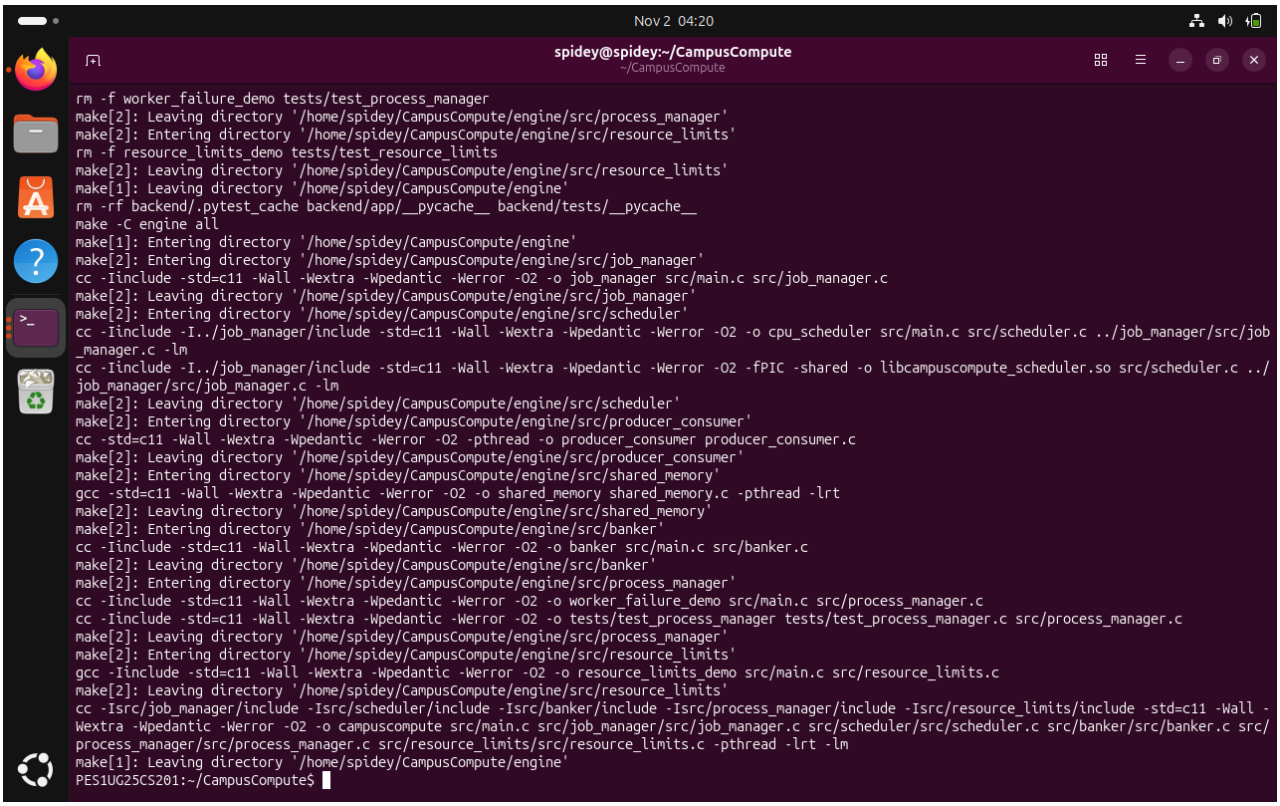
S03: Project directory listing showing the C source files, header files and Makefiles.



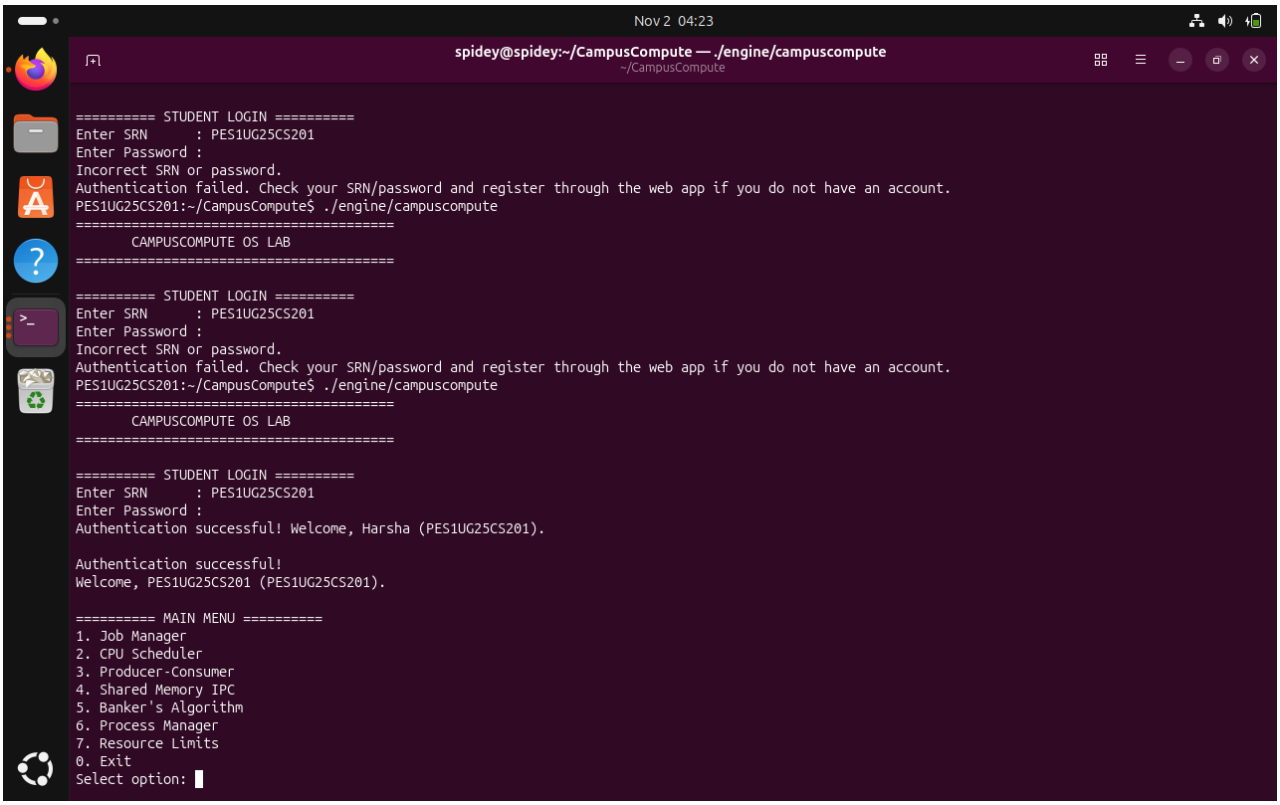
S04 – Part 1 of 2: make clean && make build starts a clean rebuild of every engine module from source.



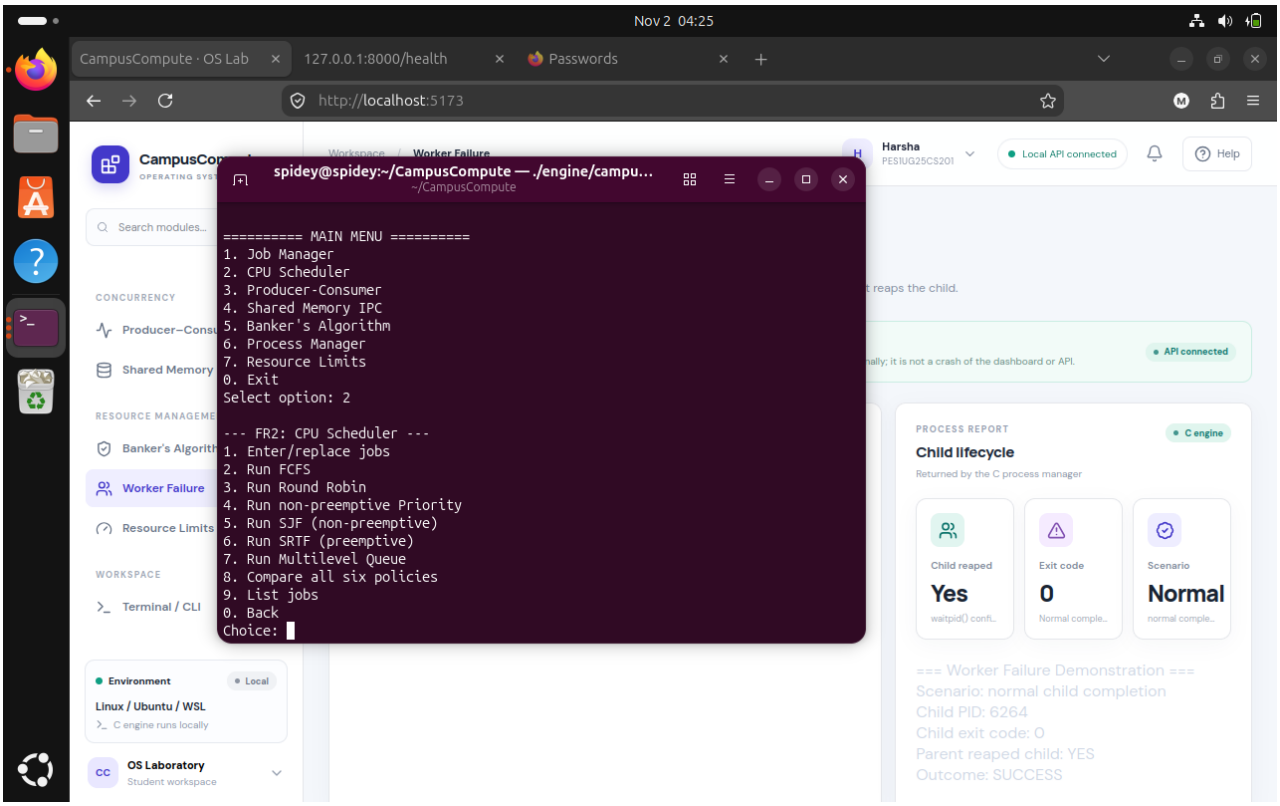
S04 – Part 2 of 2: The build completes and links the campuscompute executable without compiler errors.



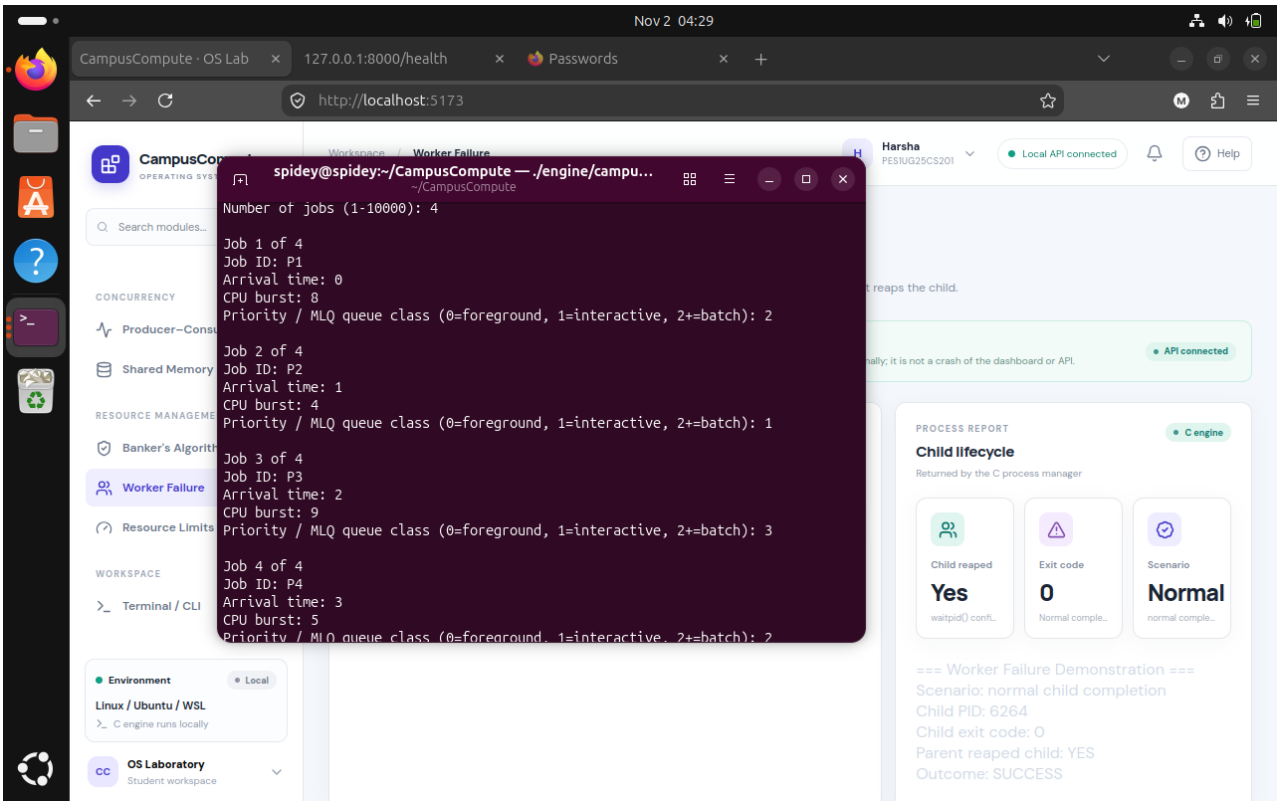
S05: The program asks for SRN login, rejects wrong credentials, then shows the main menu of available commands.



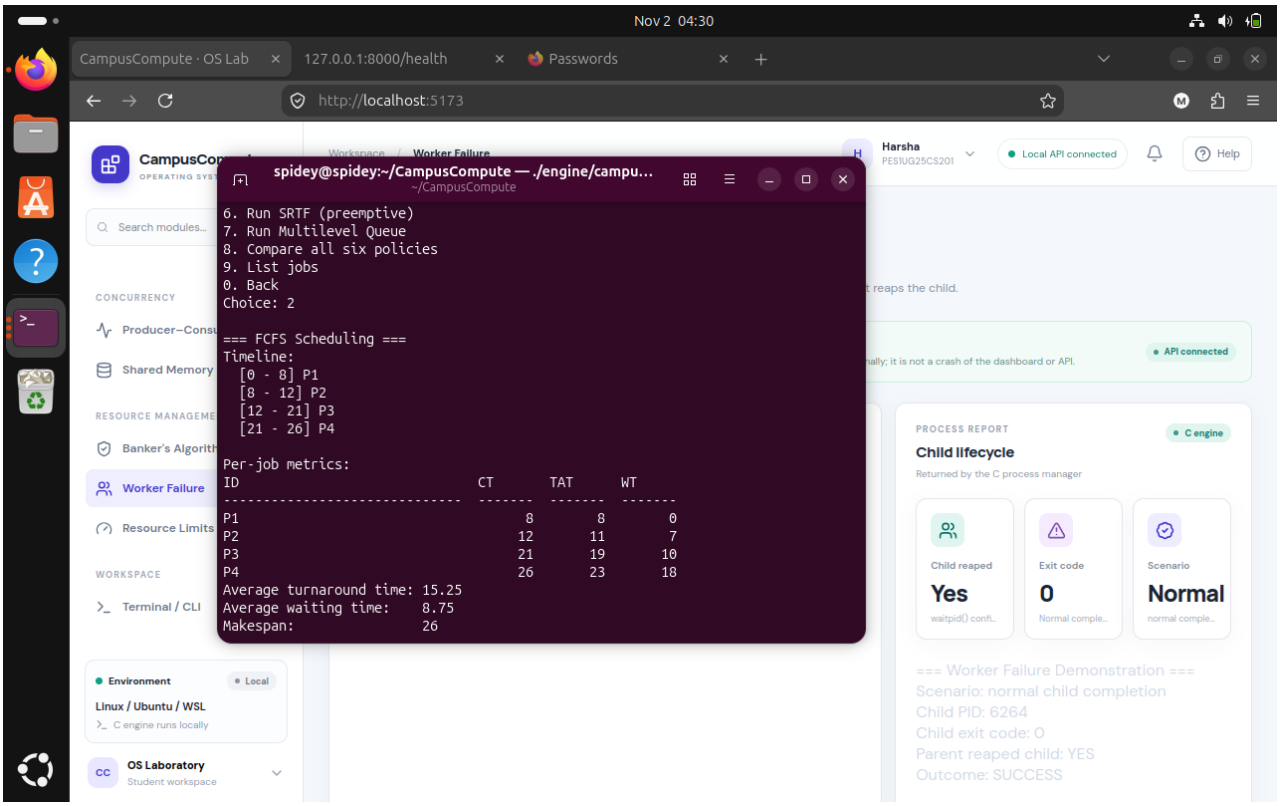
S06 – Part 1 of 3: The CPU Scheduler submenu listing the available scheduling policies and options.



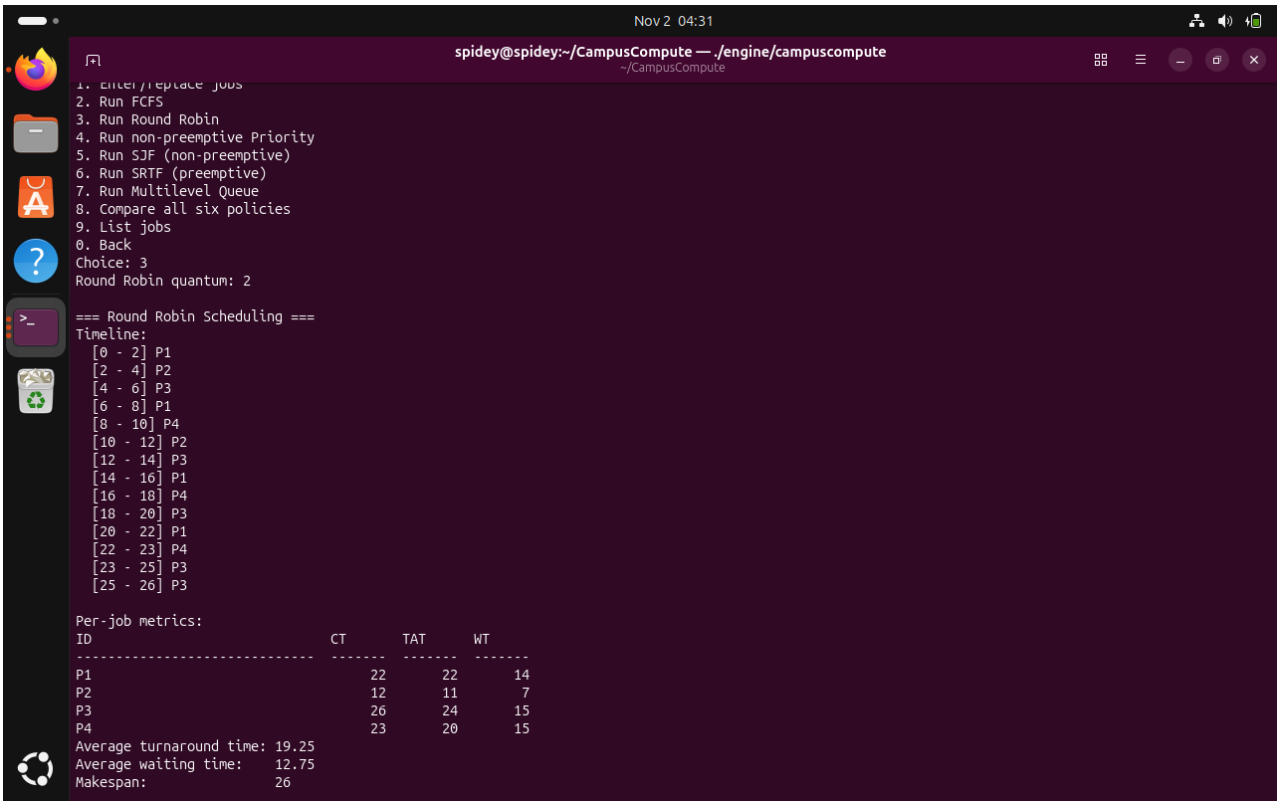
S06 – Part 2 of 3: Four jobs (P1–P4) are entered with arrival time, CPU burst and priority.



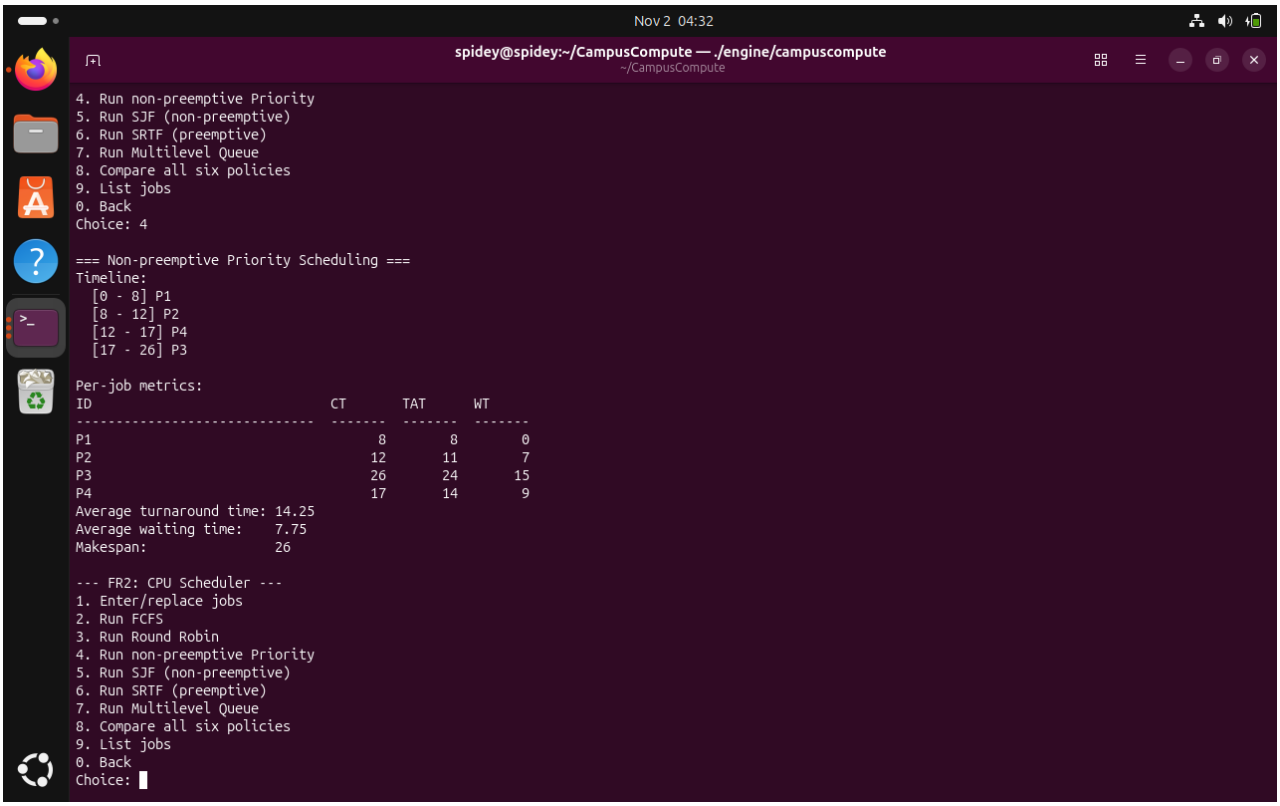
S06 – Part 3 of 3: FCFS timeline with completion, turnaround and waiting times, and the average turnaround and waiting times.



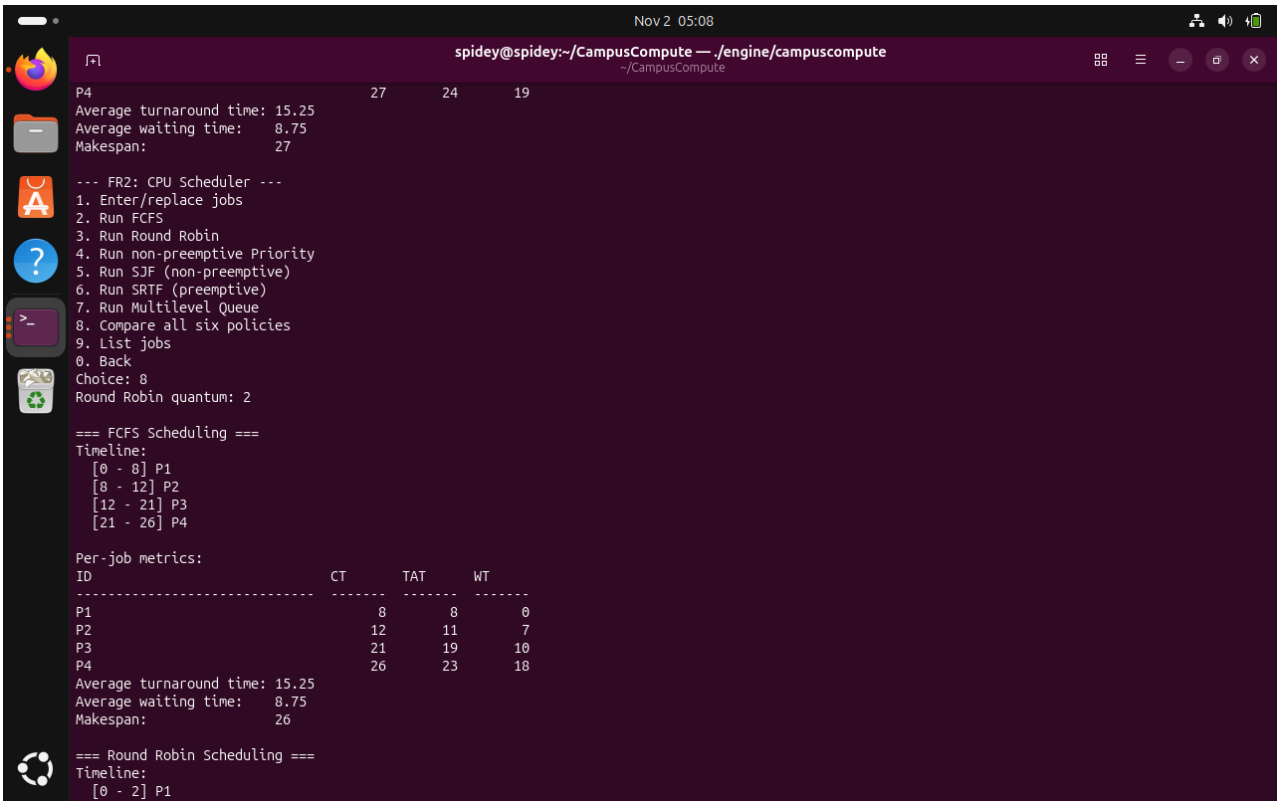
S07: Round Robin with quantum 2 showing the execution timeline and per-job scheduling metrics.



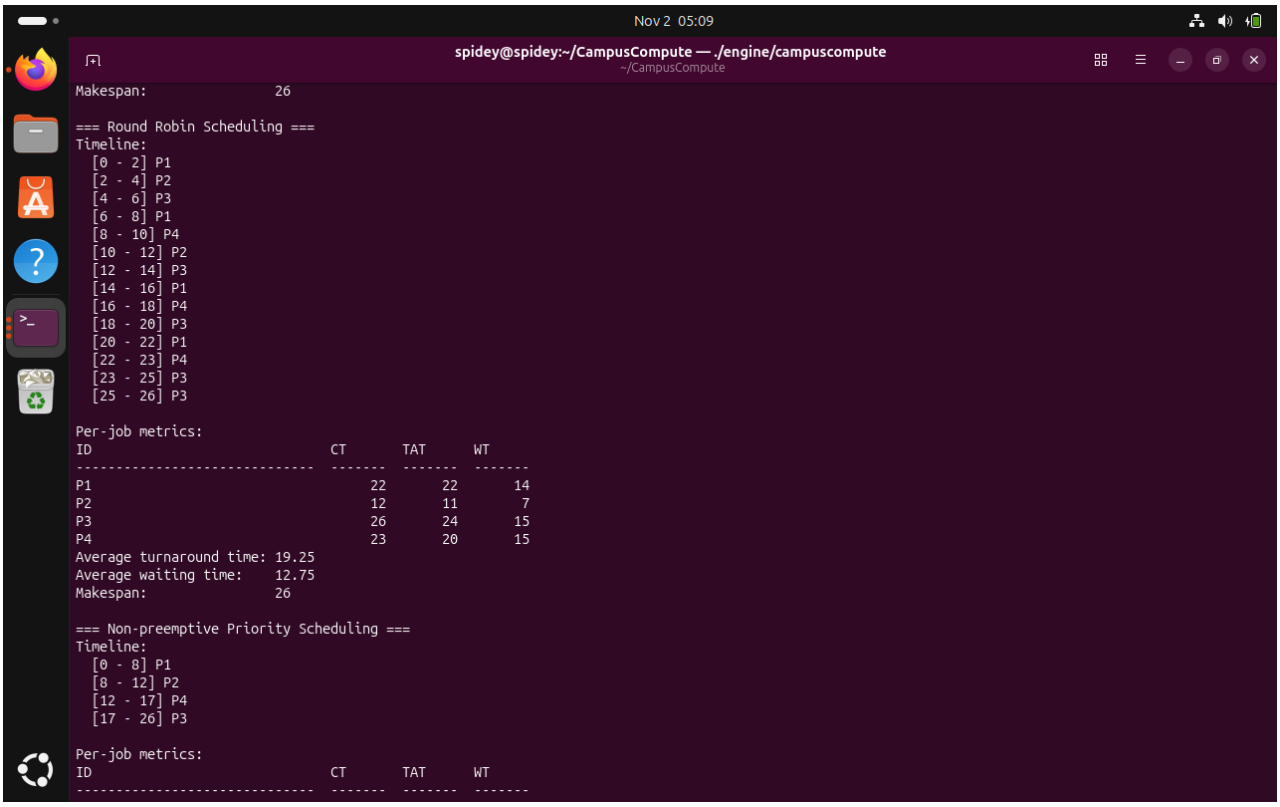
S08: Non-preemptive Priority scheduling showing the execution order and per-job scheduling metrics.



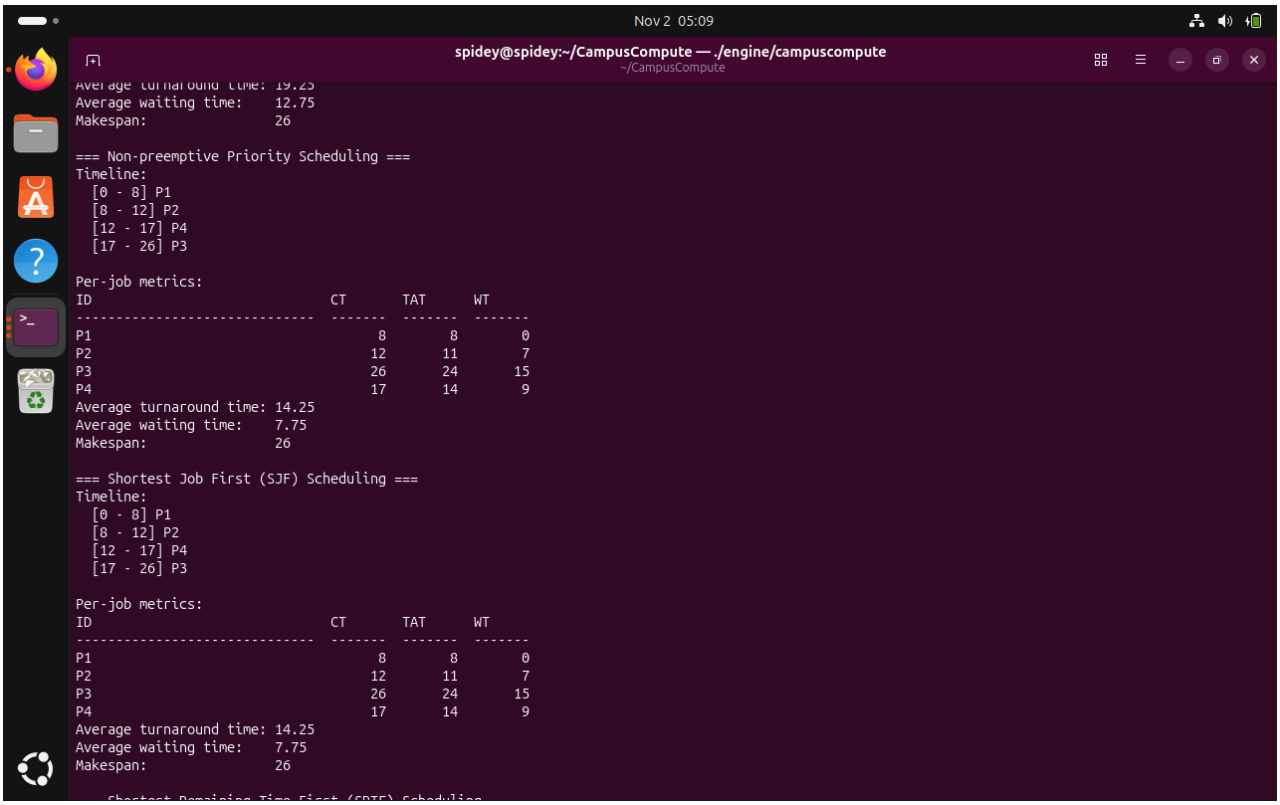
S09 – Part 1 of 5: Compare-all-policies run on the same job set, starting with the FCFS results.



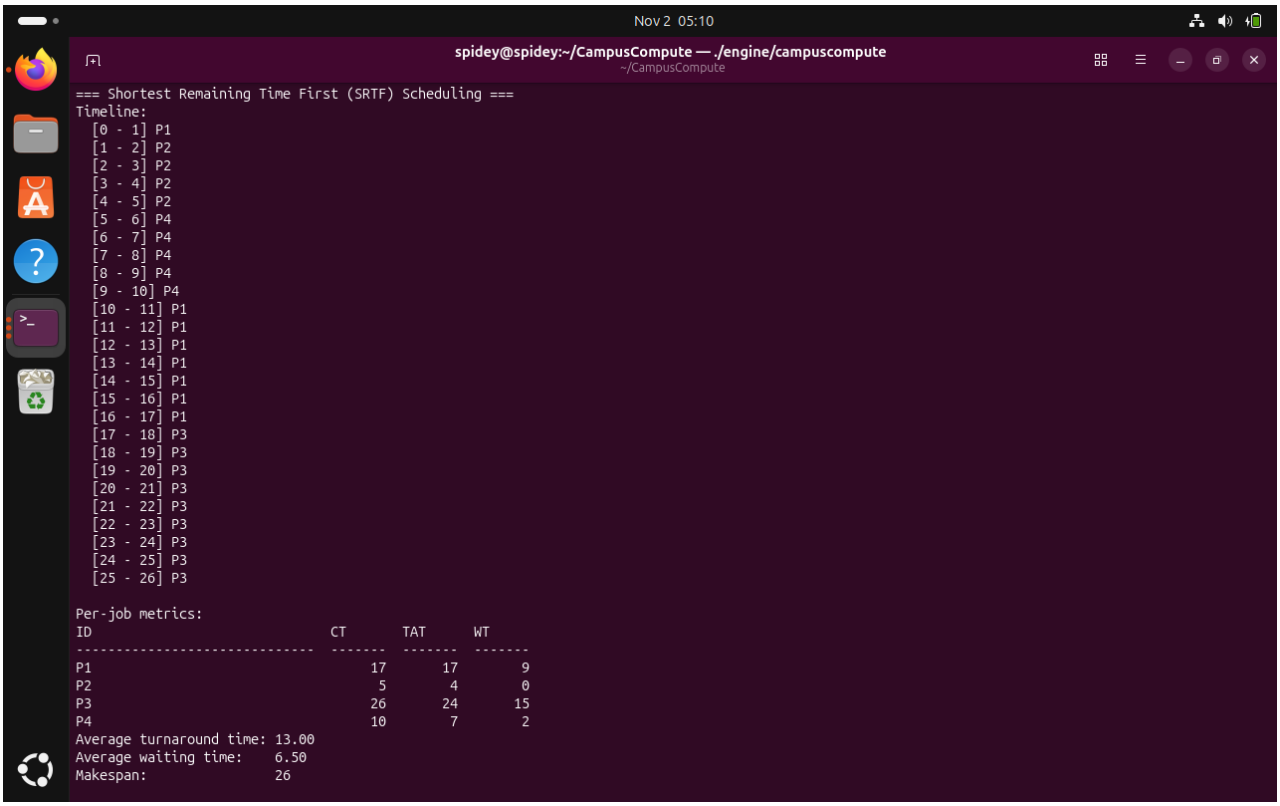
S09 – Part 2 of 5: Comparison continues with the Round Robin results and the start of Priority scheduling.



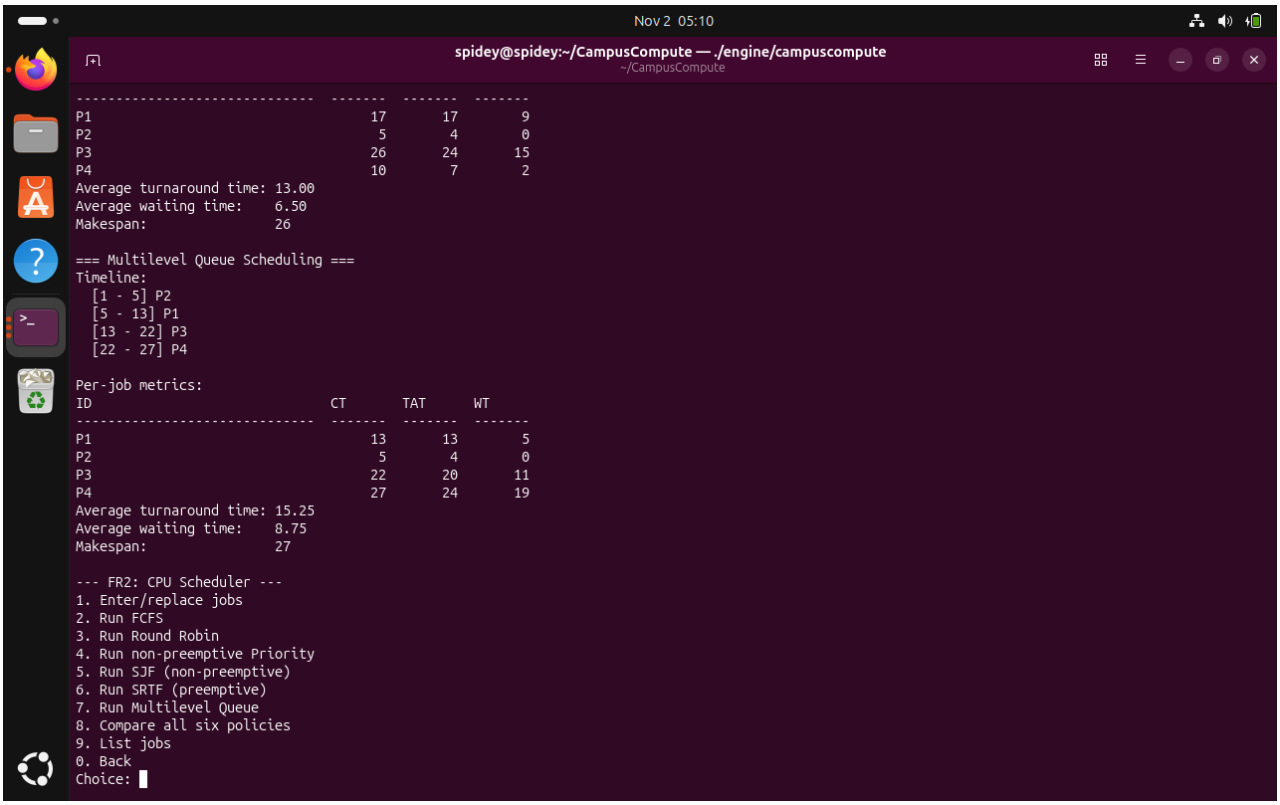
S09 – Part 3 of 5: Comparison continues with Priority and SJF results and the start of SRTF.



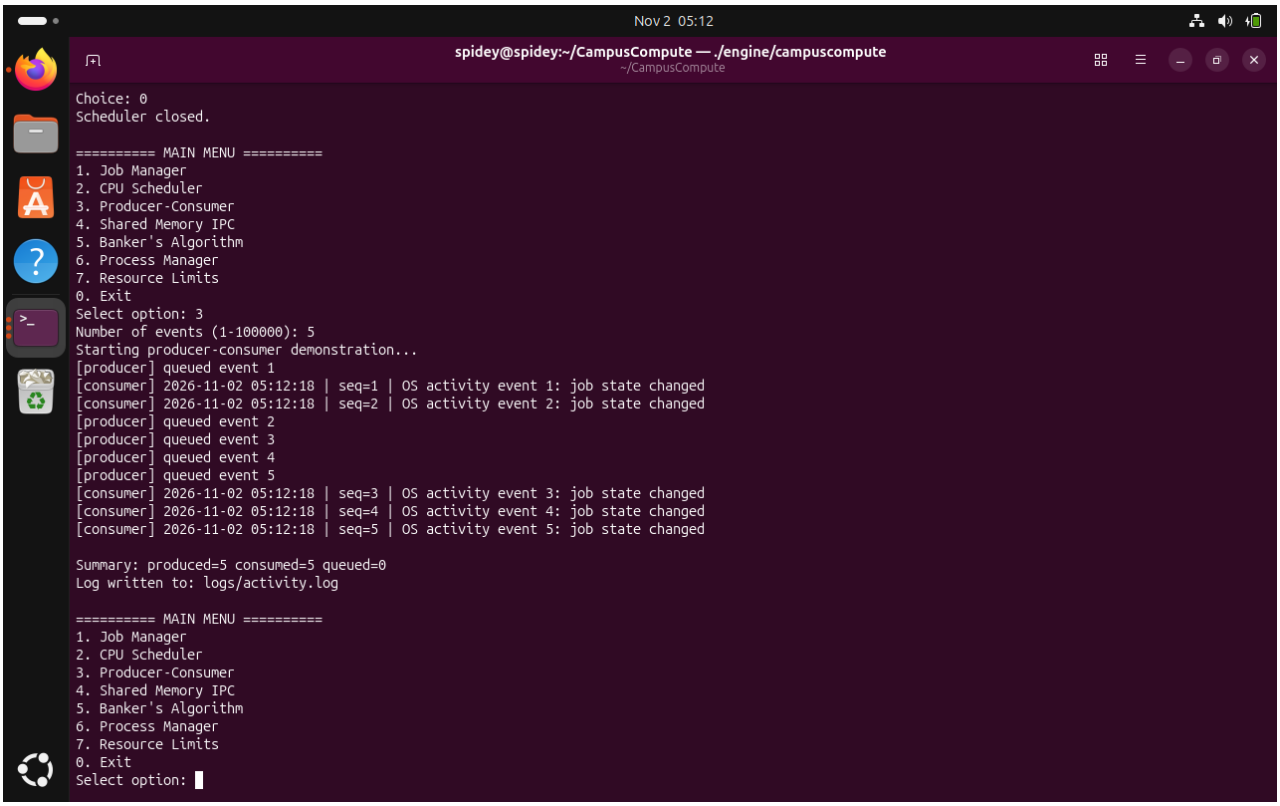
S09 – Part 4 of 5: SRTF (preemptive) timeline and per-job metrics for the same job set.



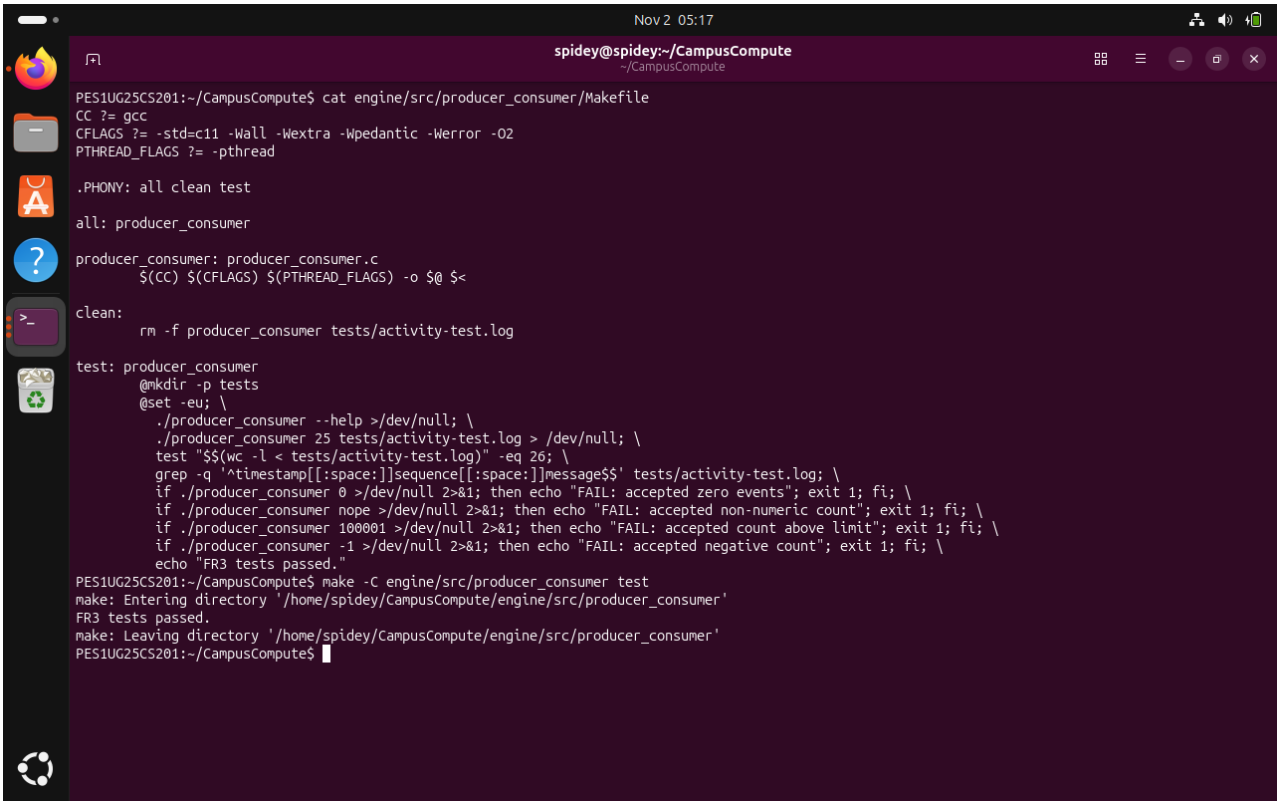
S09 – Part 5 of 5: SRTF averages followed by the Multilevel Queue results, completing the policy comparison.



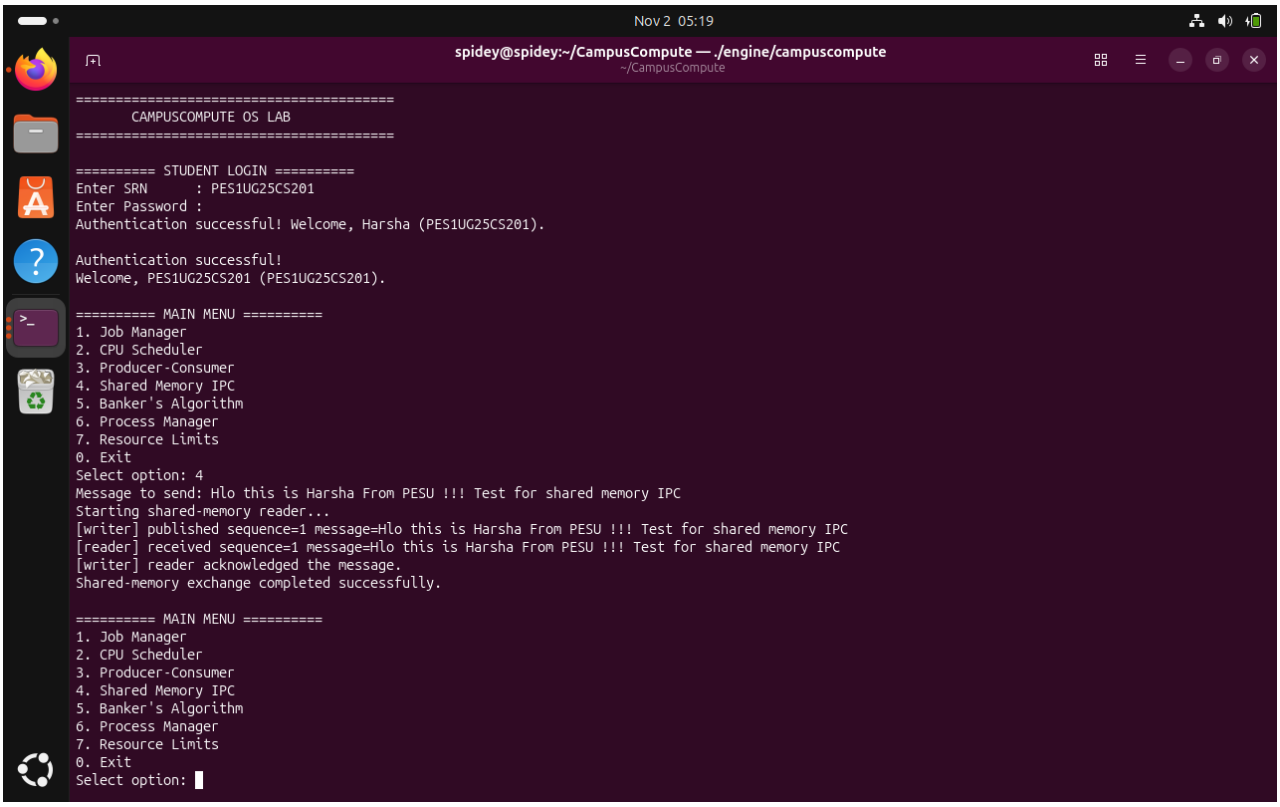
S10: Producer-consumer demonstration with 5 events: produced=5, consumed=5, queued=0, log written to logs/activity.log.



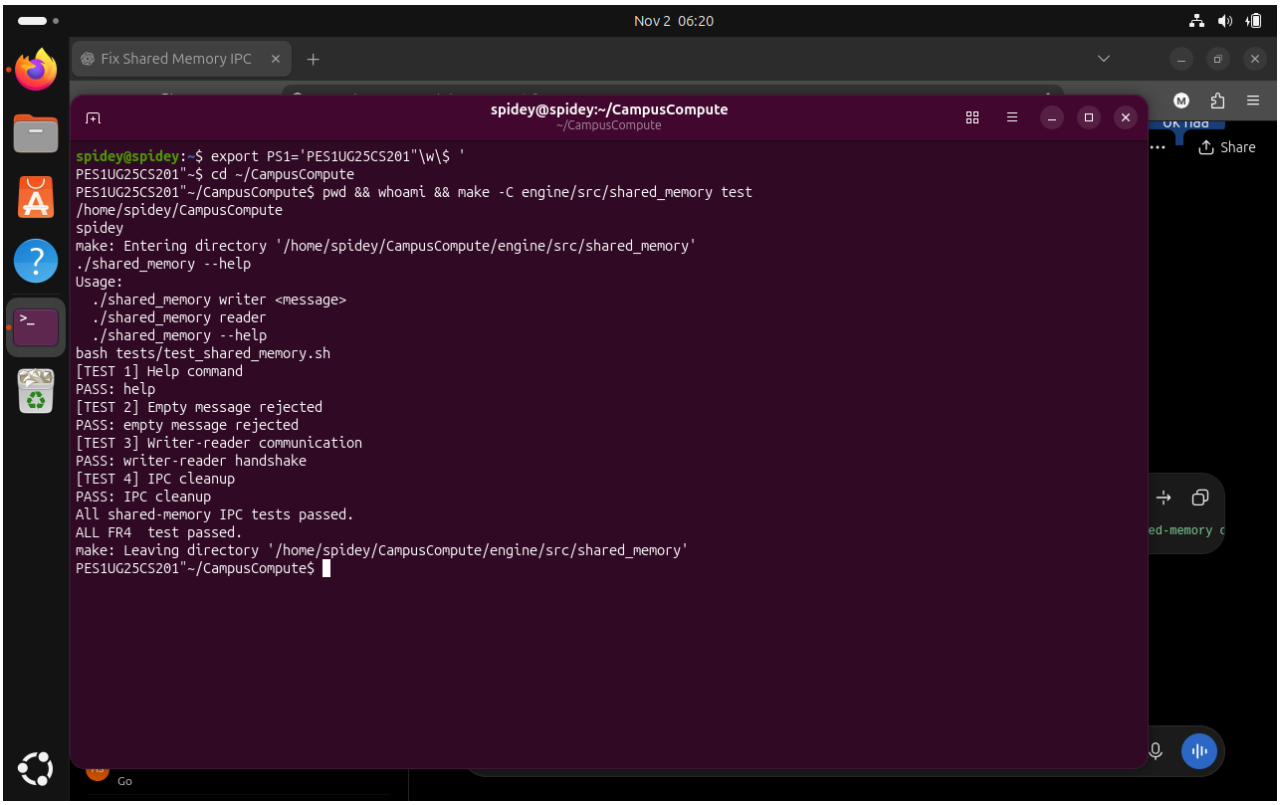
S11: Producer-consumer Makefile test run passes (FR3 tests passed), showing synchronized operation and clean thread termination.



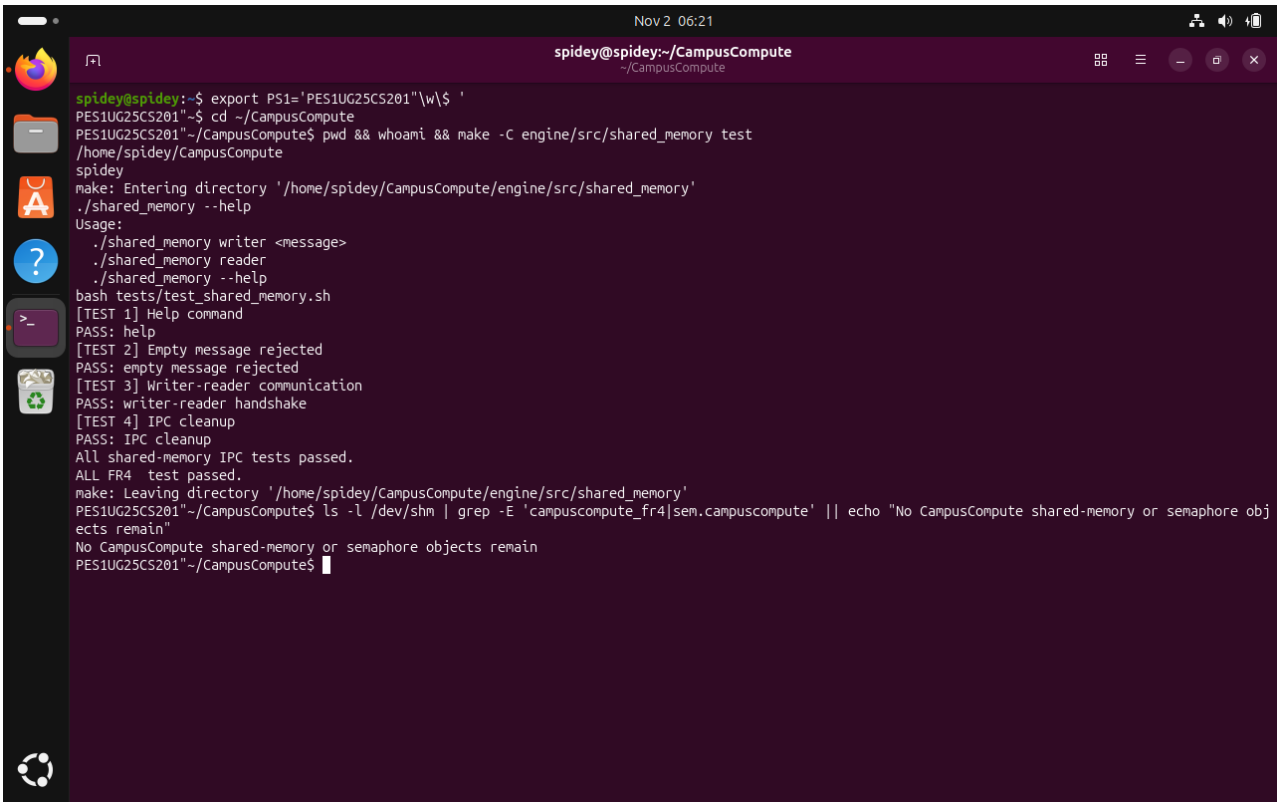
S12: Writer and reader processes exchange a message through POSIX shared memory.



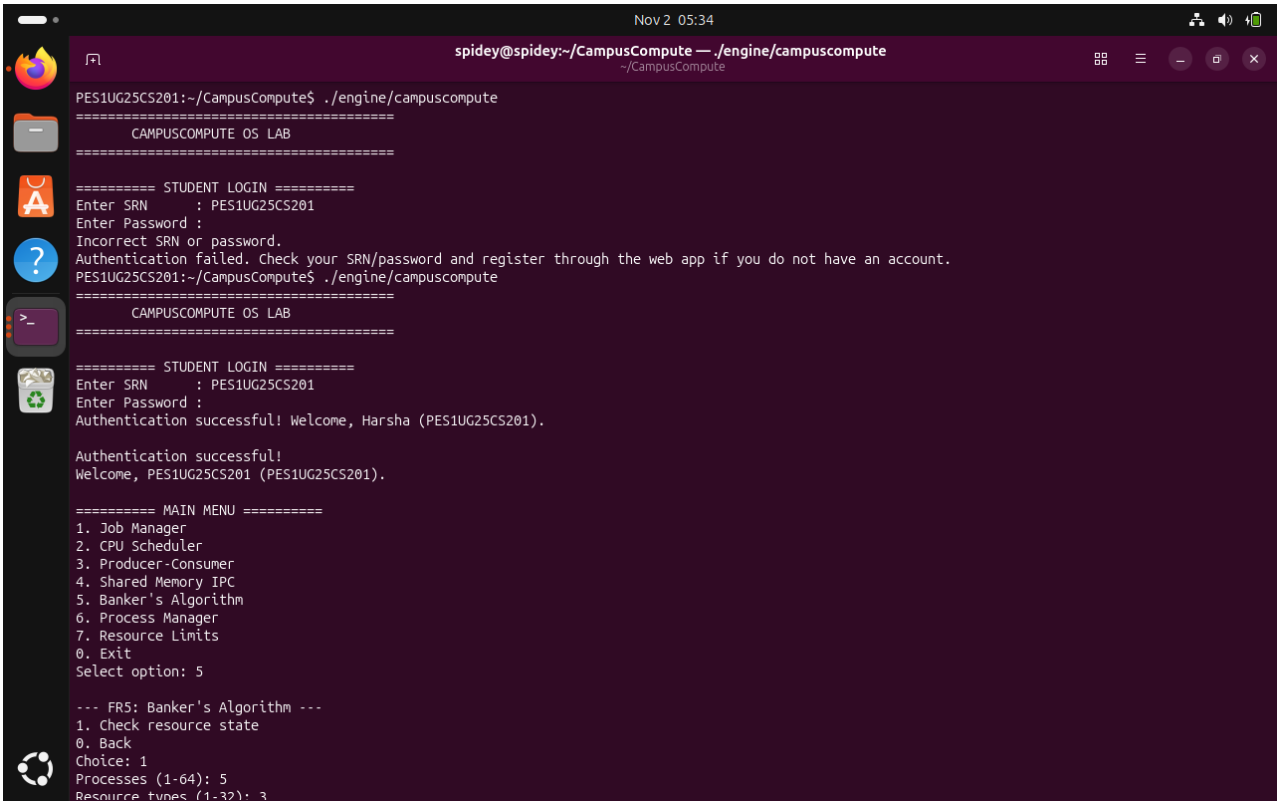
S13 – Part 1 of 2: Shared-memory test suite passes, including the semaphore-coordinated writer–reader handshake.



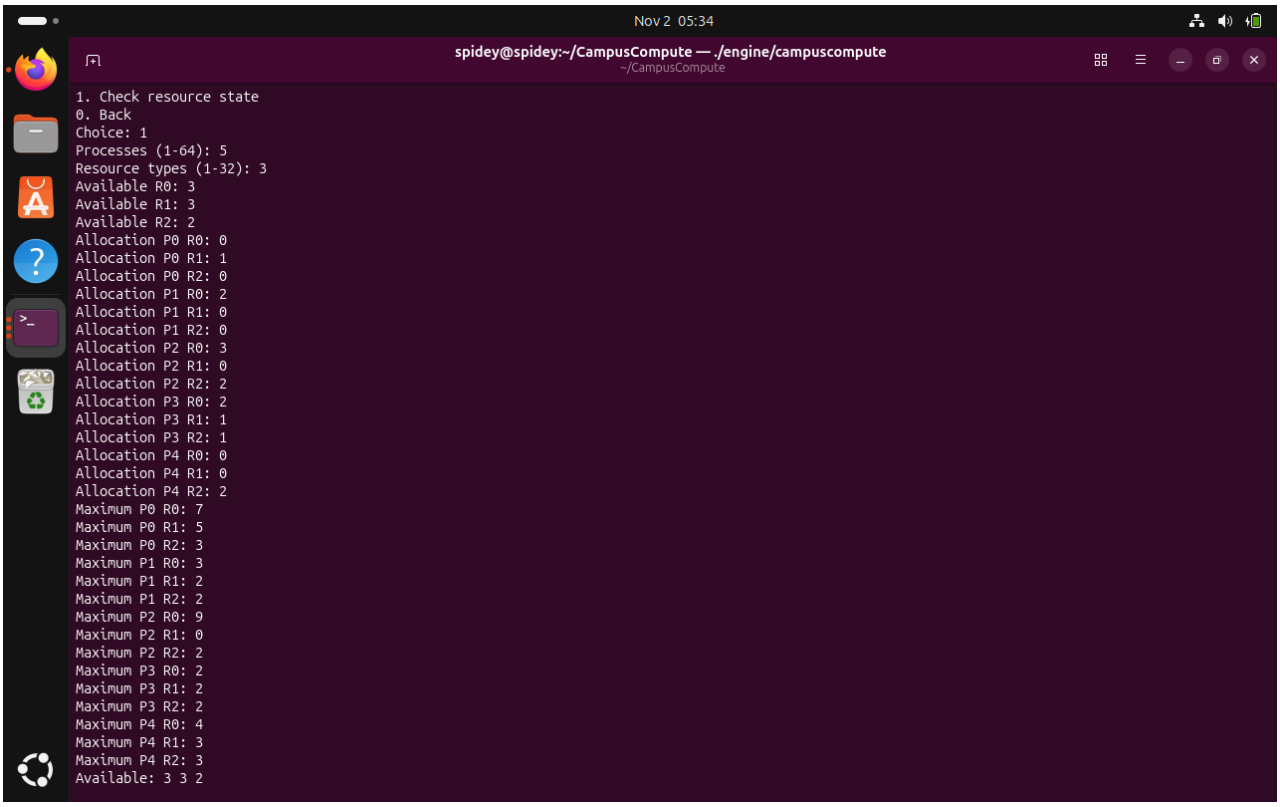
S13 – Part 2 of 2: After the tests, no CampusCompute shared-memory or semaphore objects remain in /dev/shm.



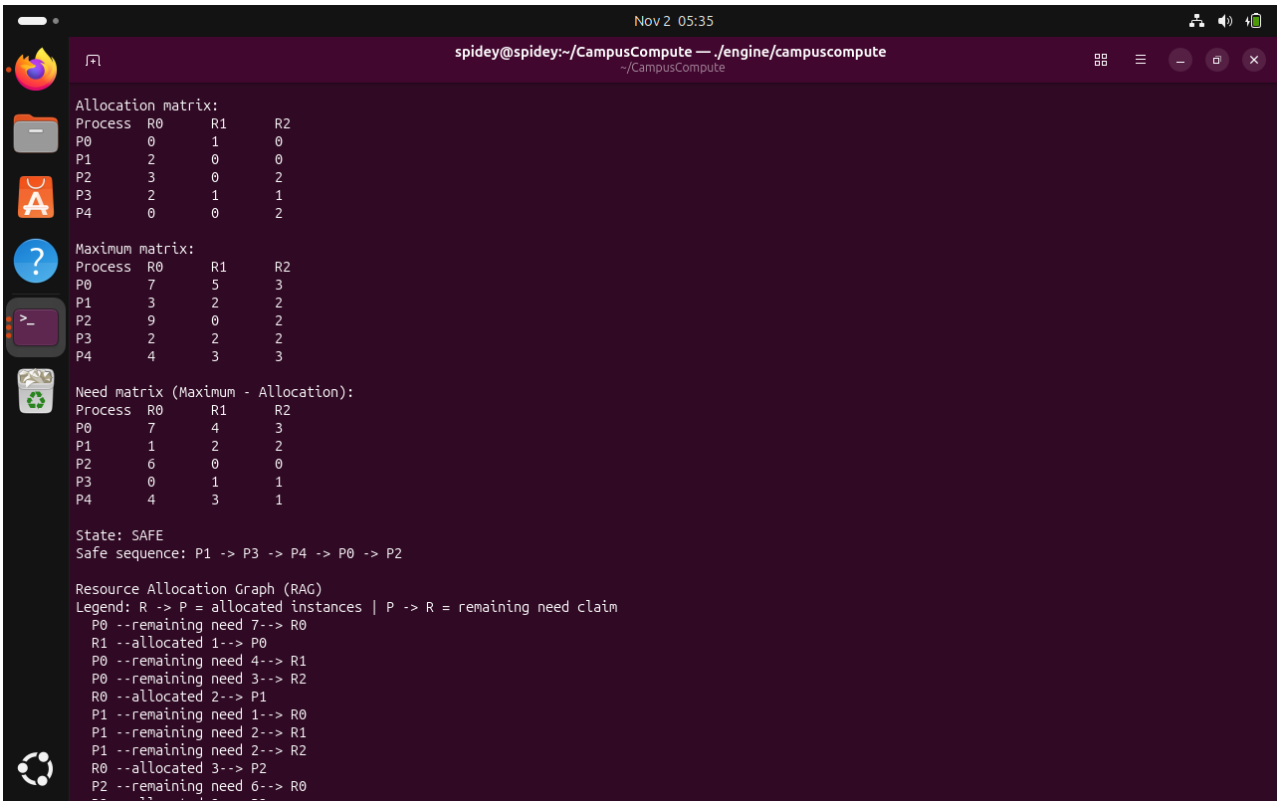
S14 – Part 1 of 4: Banker's Algorithm menu opened and the number of processes and resource types entered for the safe-state test.



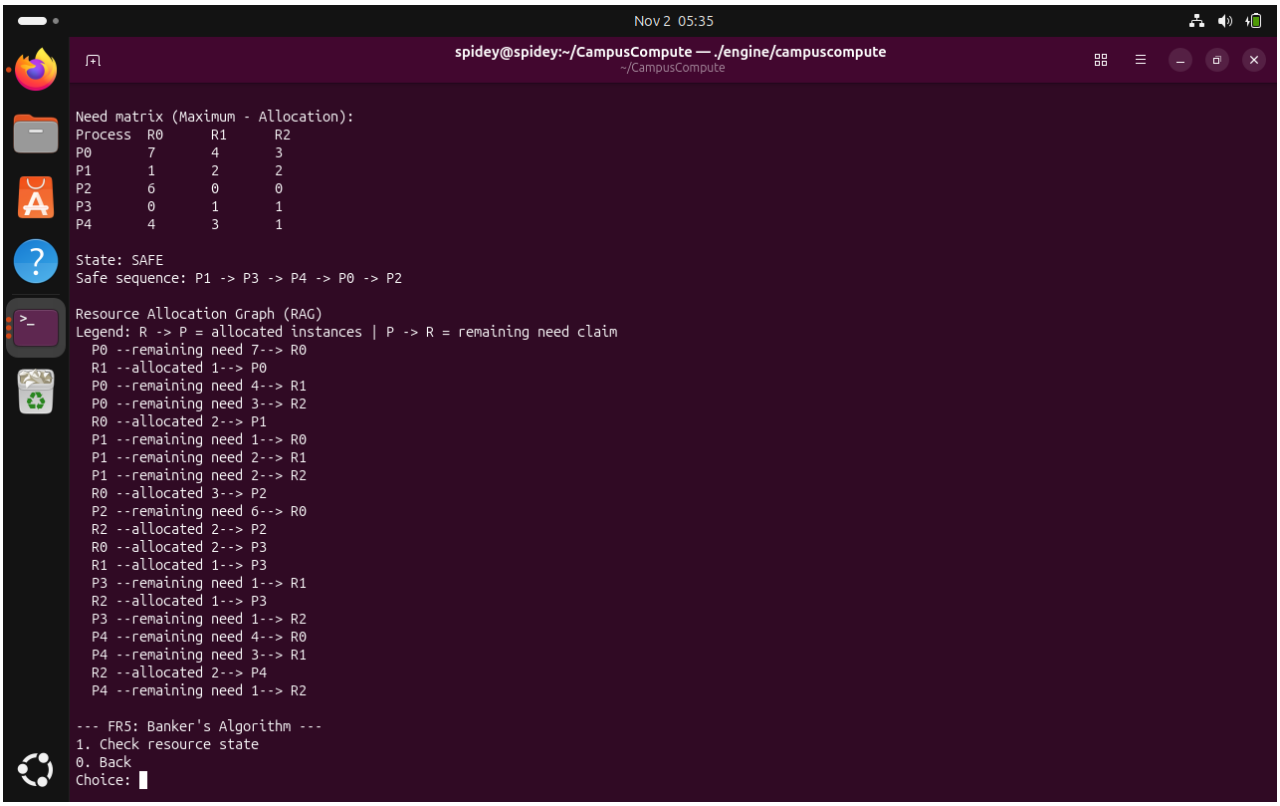
S14 – Part 2 of 4: Available, Allocation and Maximum values entered for the safe-state test.



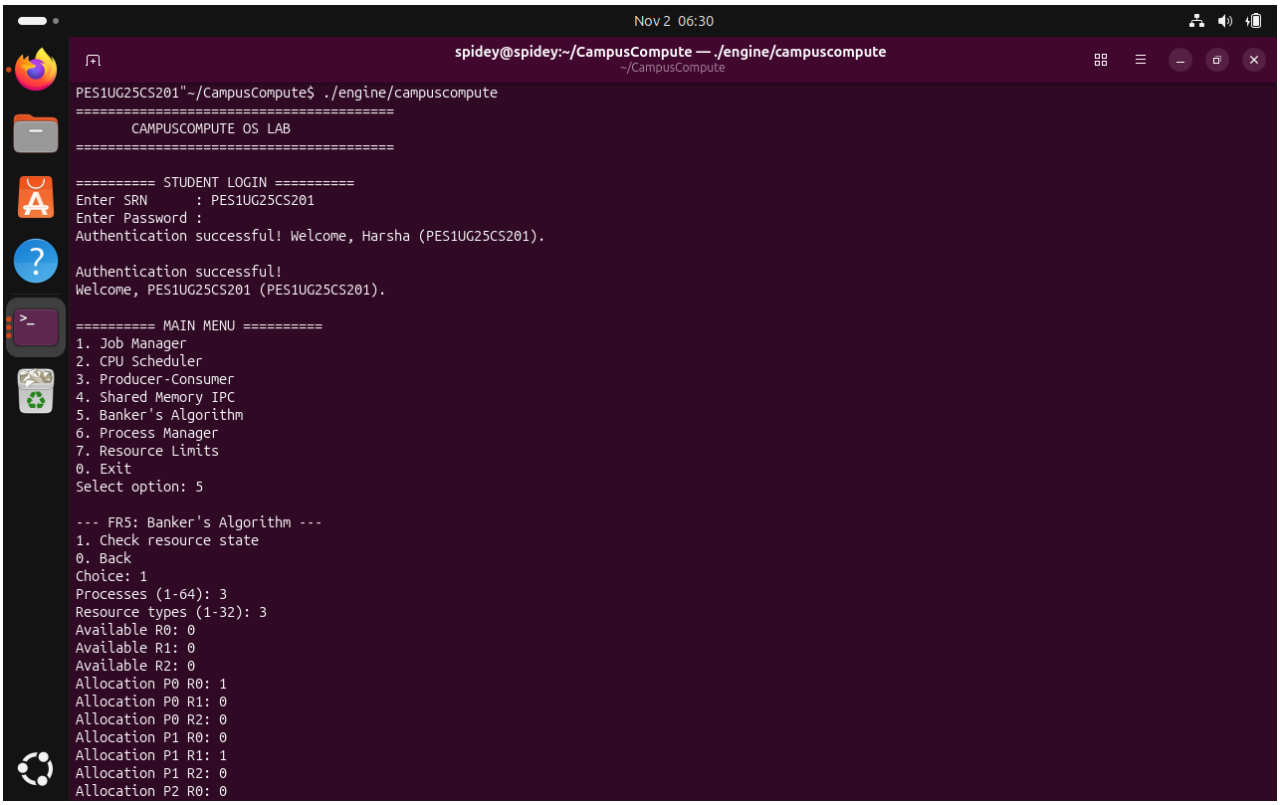
S14 – Part 3 of 4: Allocation, Maximum and Need matrices shown; state is SAFE with safe sequence P1 -> P3 -> P4 -> P0 -> P2.



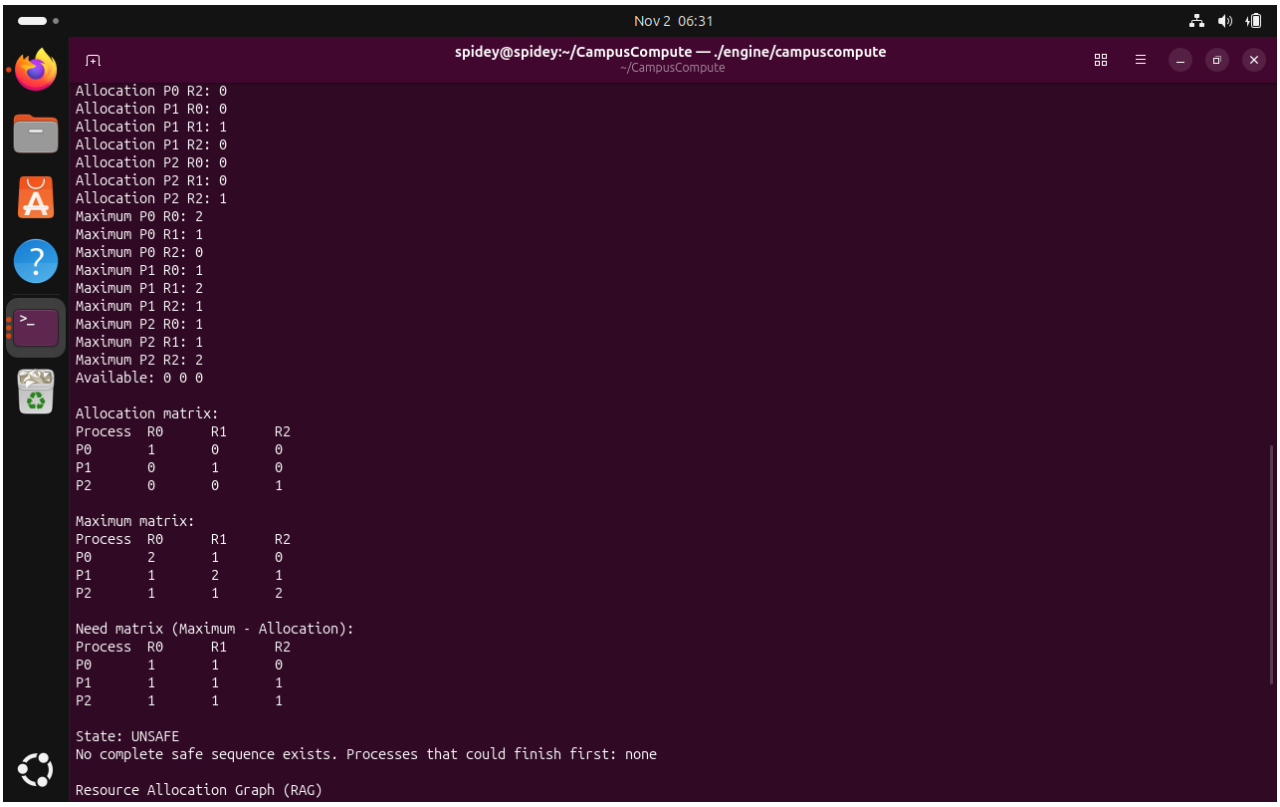
S14 – Part 4 of 4: SAFE result and safe sequence repeated, followed by the resource allocation graph.



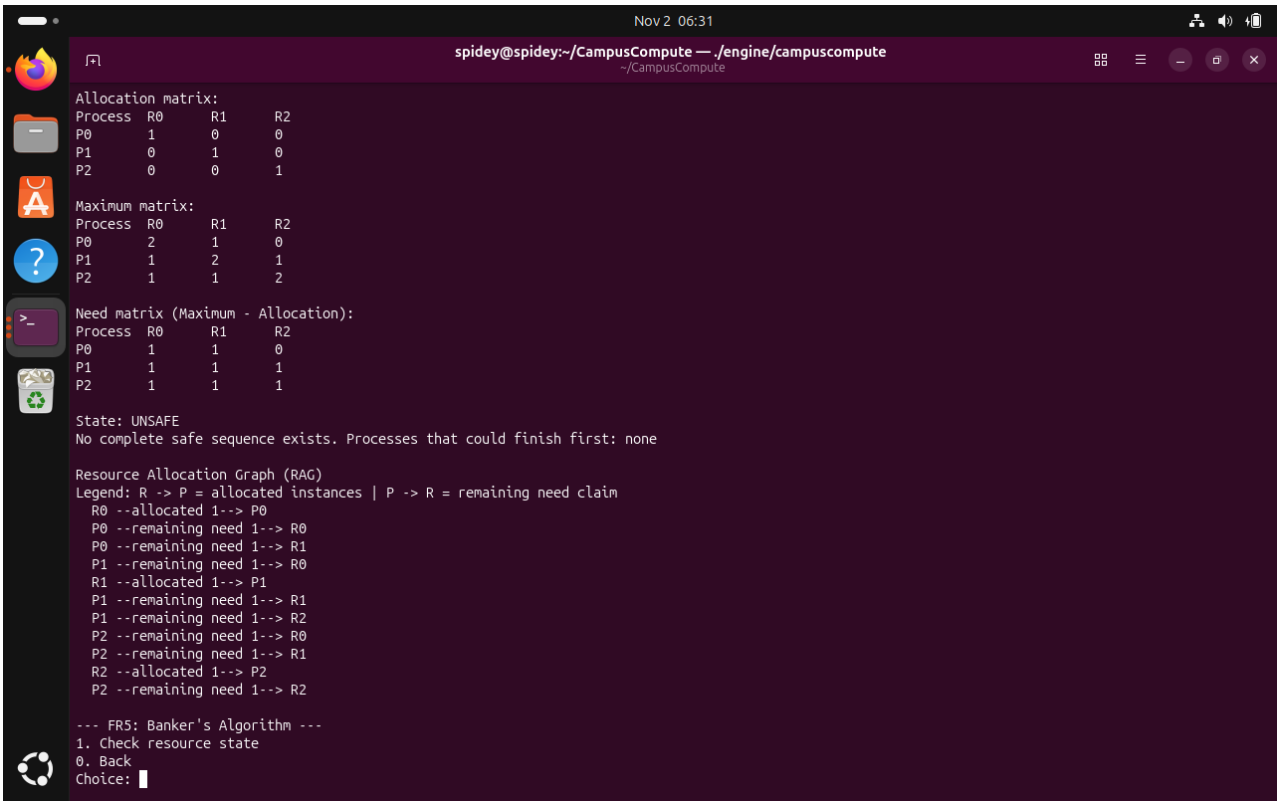
S15 – Part 1 of 4: Unsafe-state test: 3 processes and 3 resource types entered with Available set to 0 0 0.



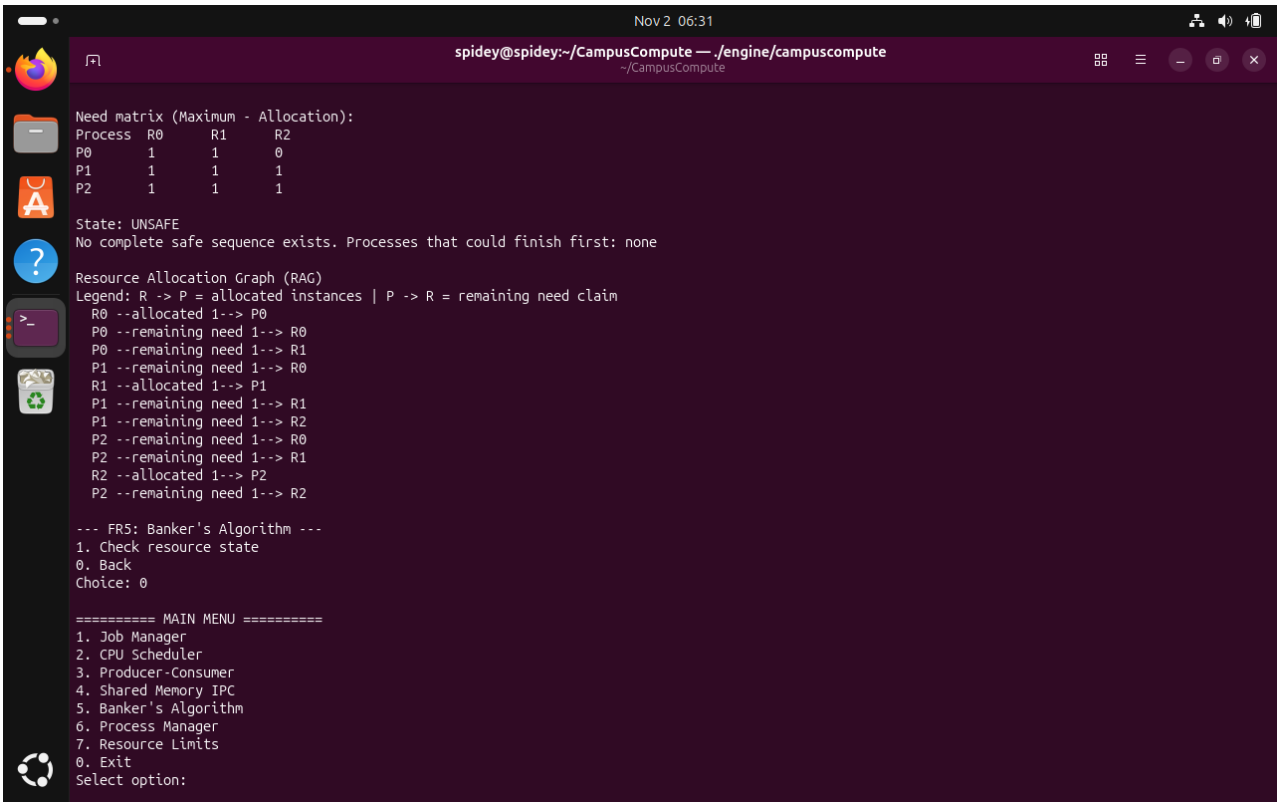
S15 – Part 2 of 4: Allocation and Maximum values entered; the algorithm reports State: UNSAFE.



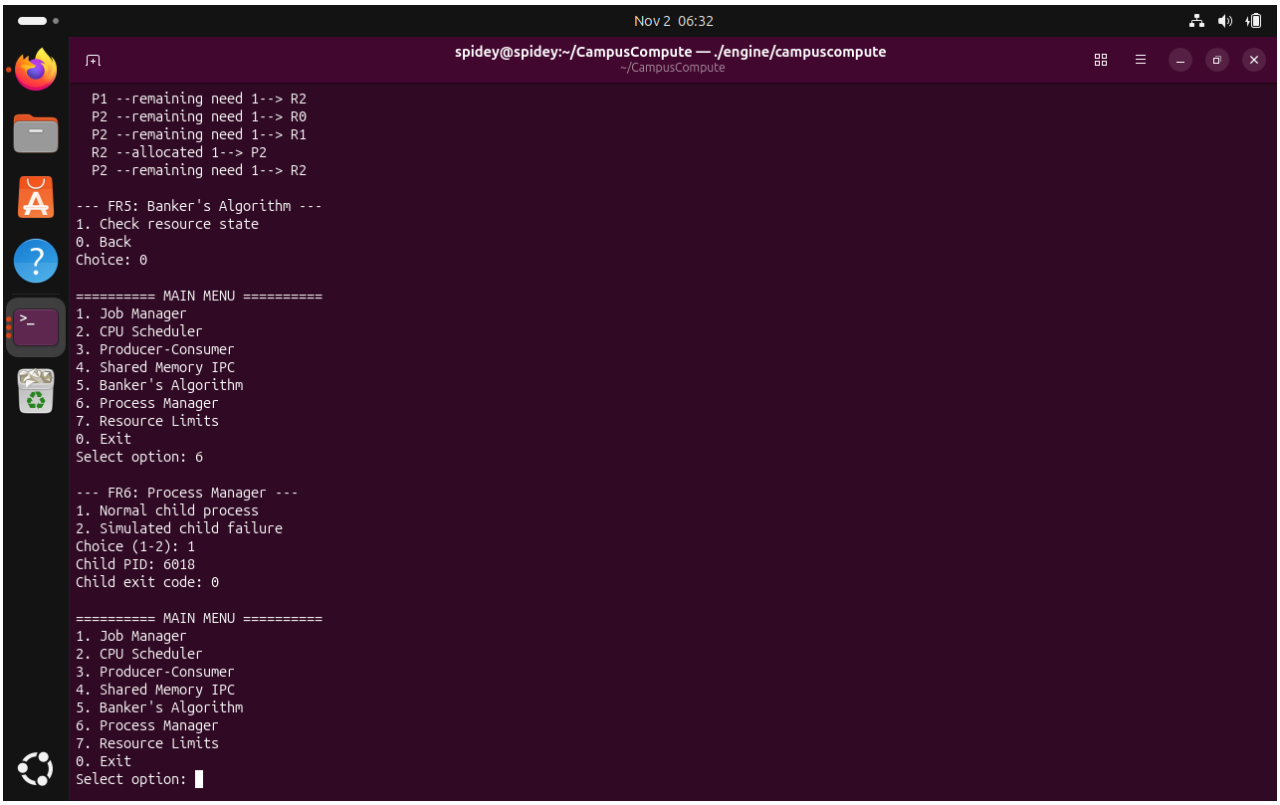
S15 – Part 3 of 4: Matrices, UNSAFE state (no complete safe sequence) and the resource allocation graph.



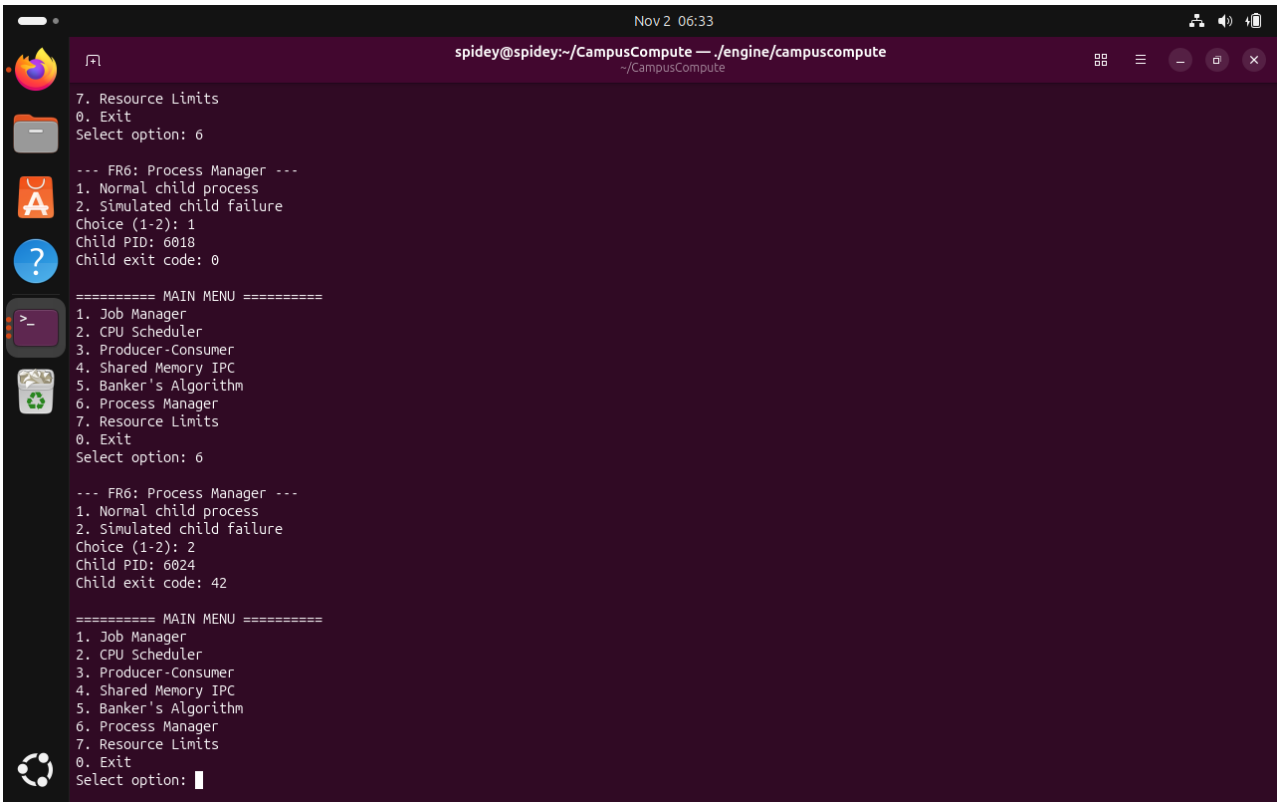
S15 – Part 4 of 4: UNSAFE result and graph shown again before returning to the main menu.



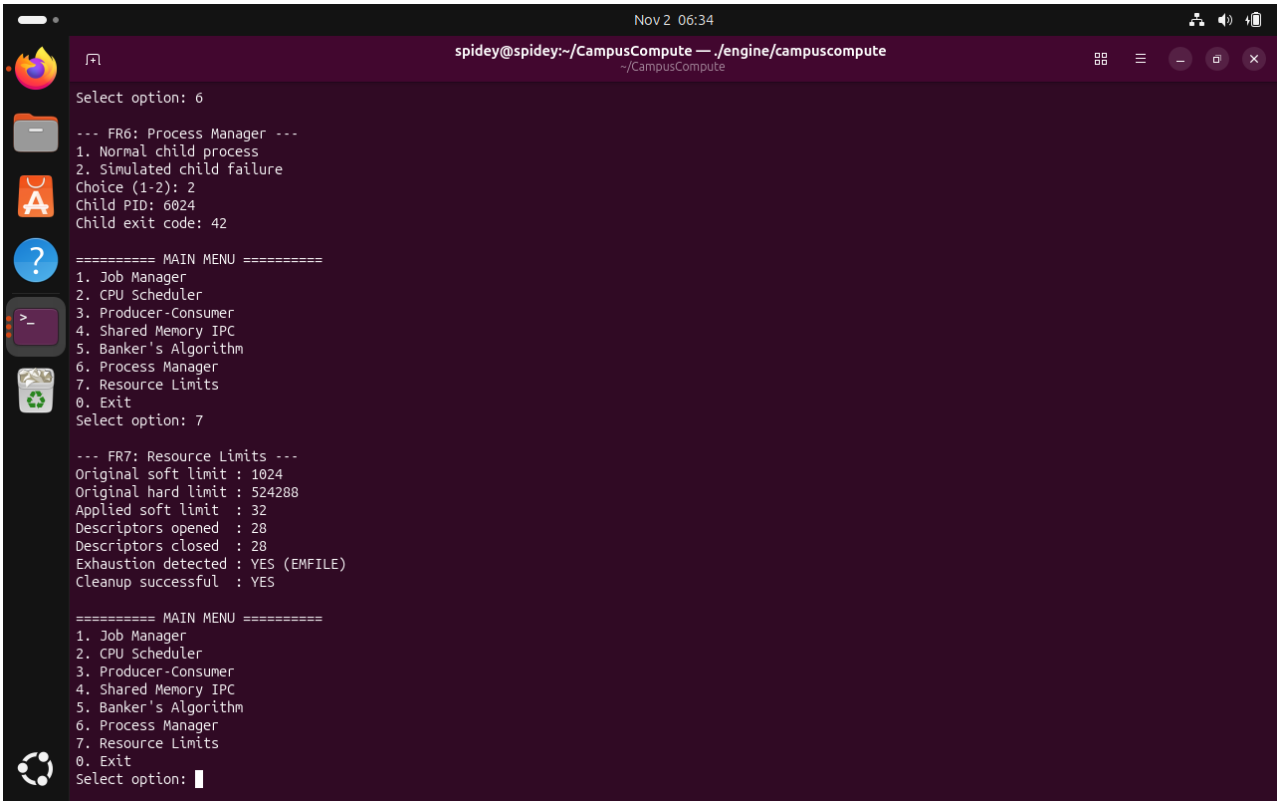
S16: Process Manager creates a normal child process (PID 6018) that completes with exit code 0.



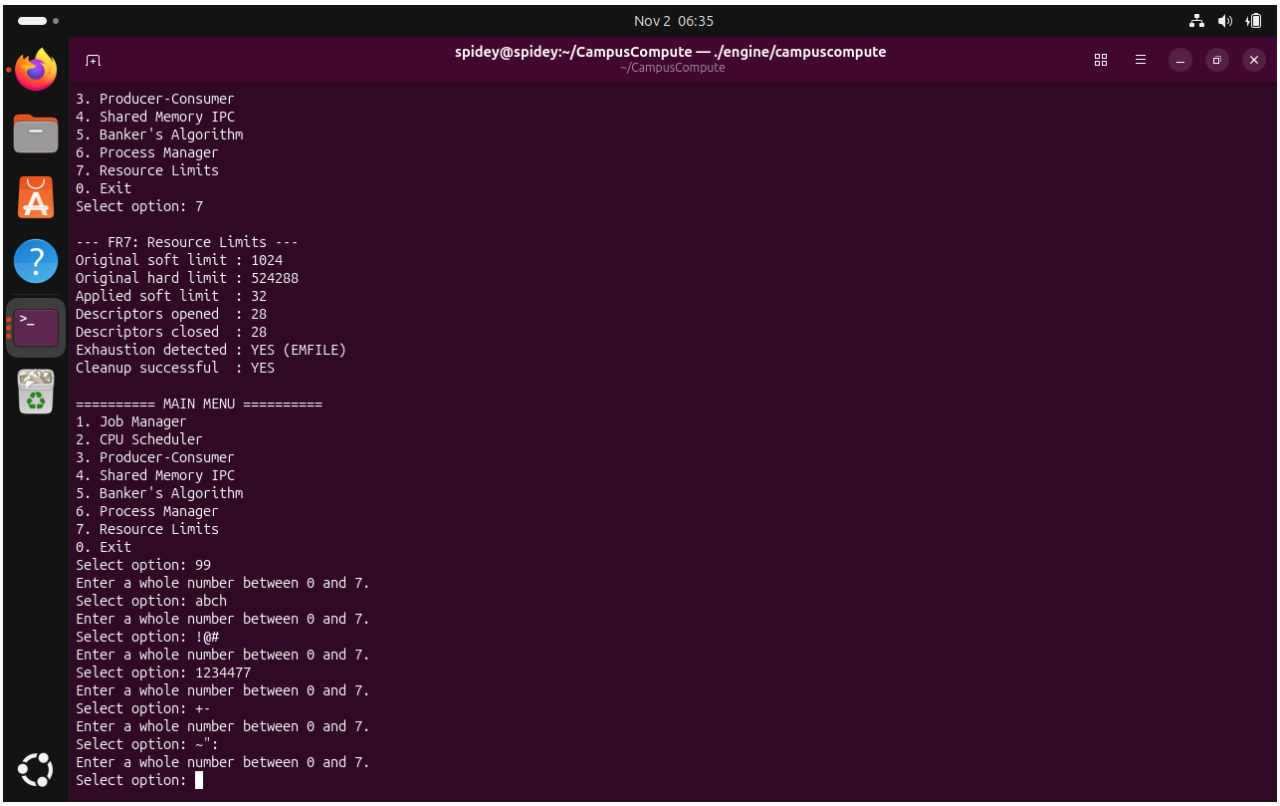
S17: Simulated worker failure: child PID 6024 terminates with exit code 42, detected by the parent (simulated, not a real OS failure).



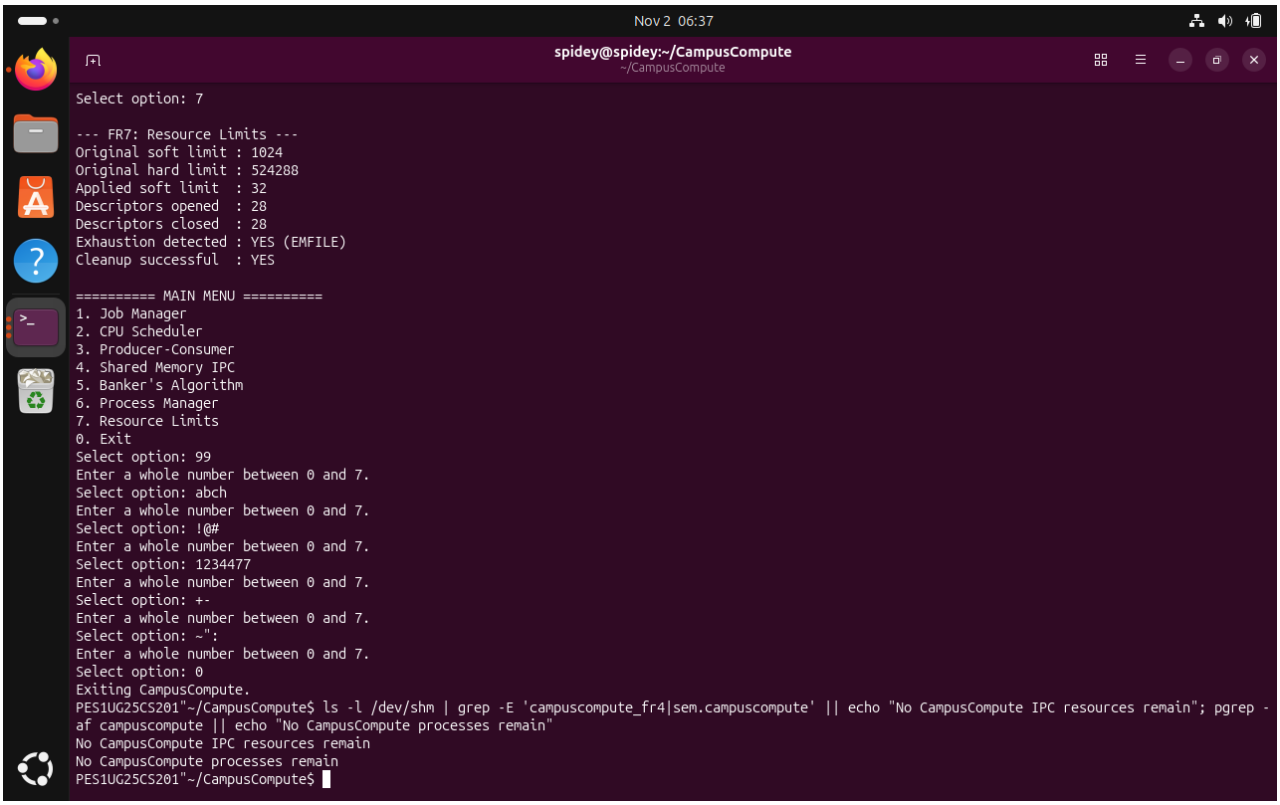
S18: Soft limit lowered to 32 descriptors; 28 opened before exhaustion was detected (EMFILE) and cleanup succeeded.



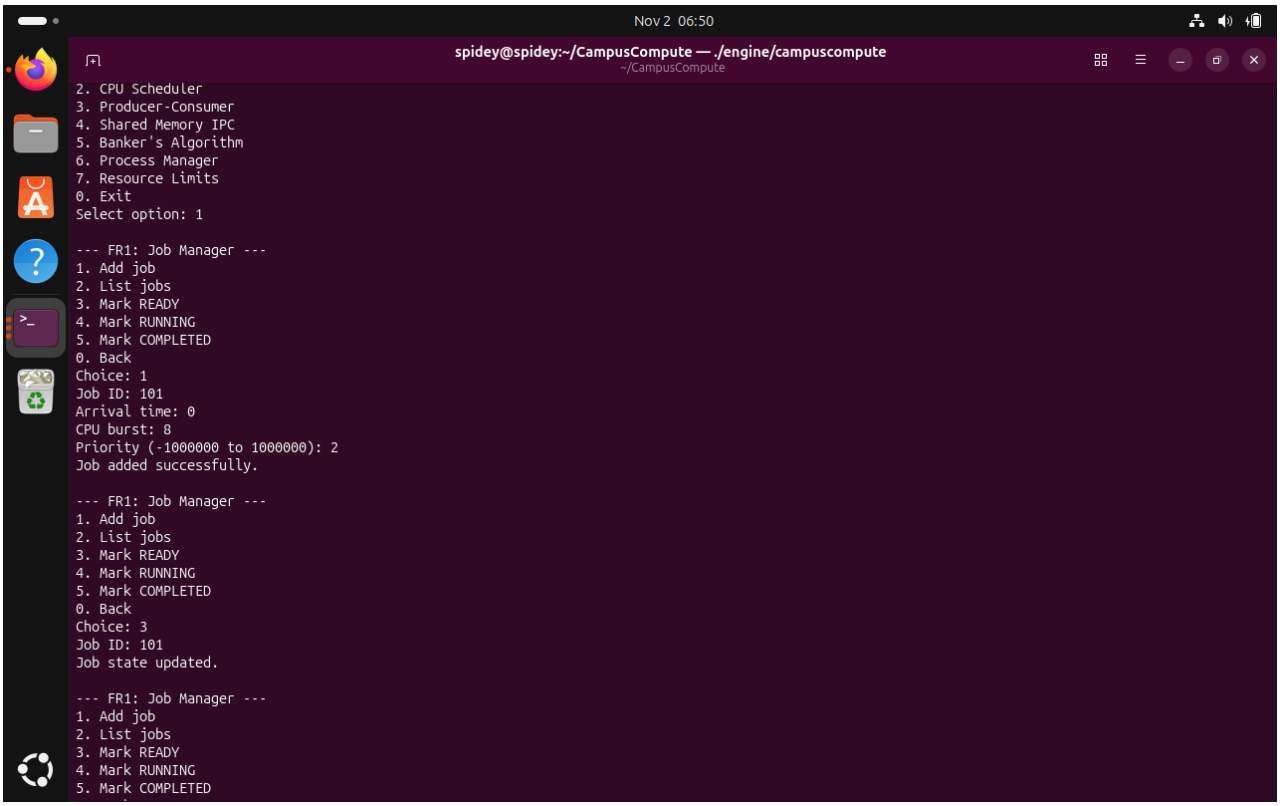
S19: Invalid menu inputs are rejected with the message “Enter a whole number between 0 and 7.”



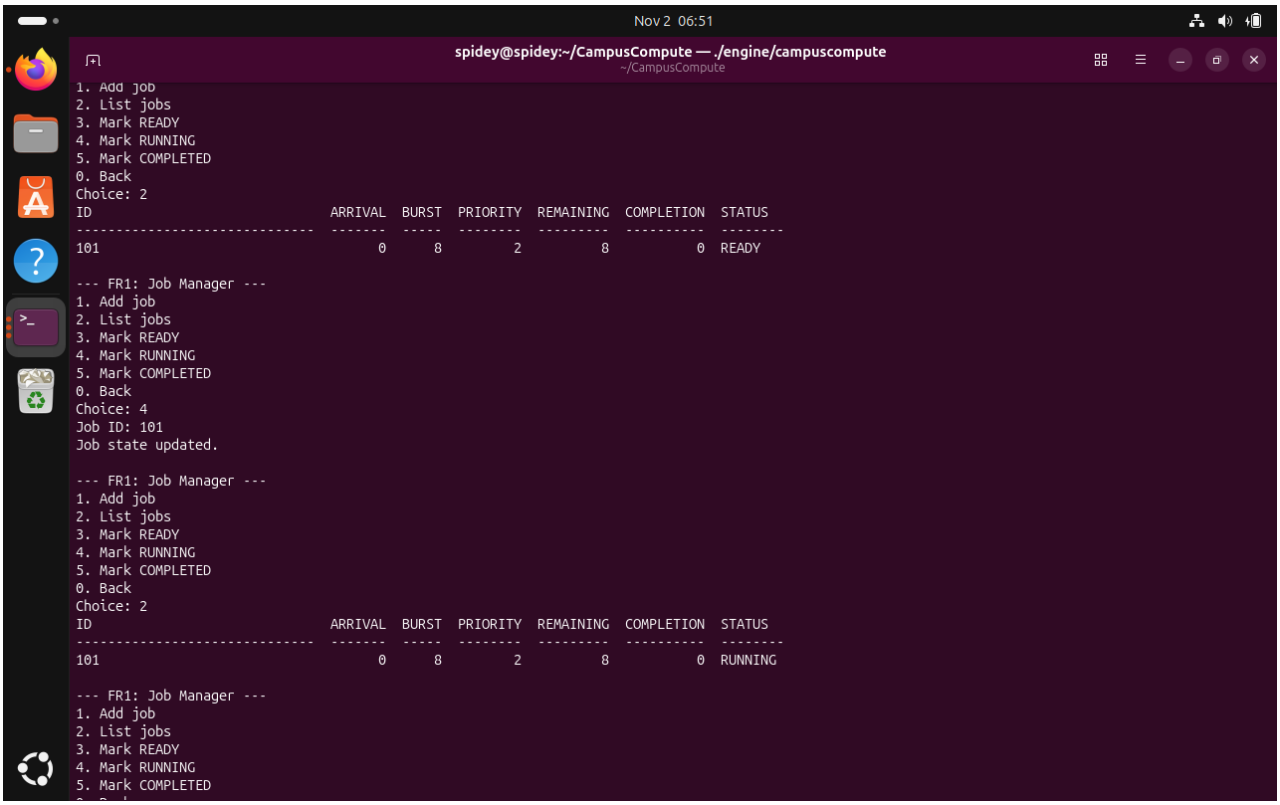
S20: After exiting, no CampusCompute IPC resources or processes remain.



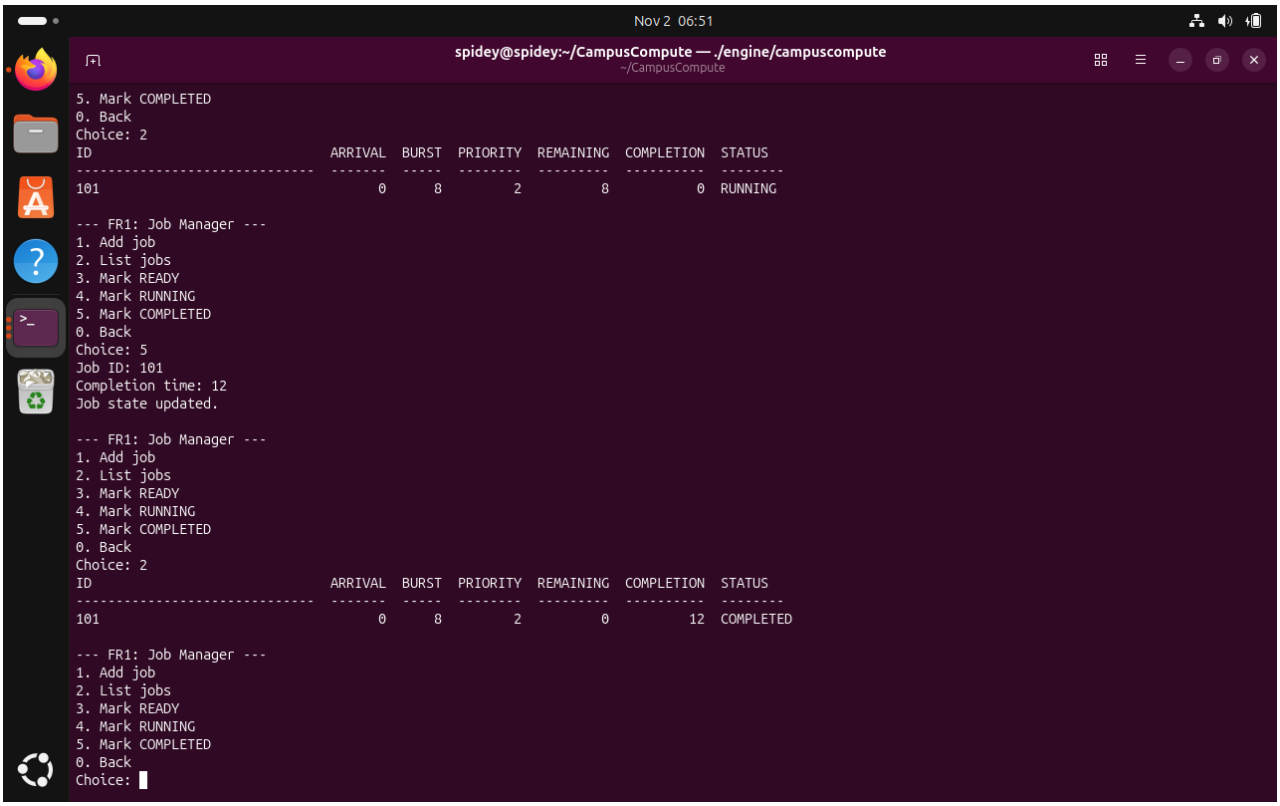
S21 – Part 1 of 4: Integrated run: job 101 is added in the Job Manager and marked READY.



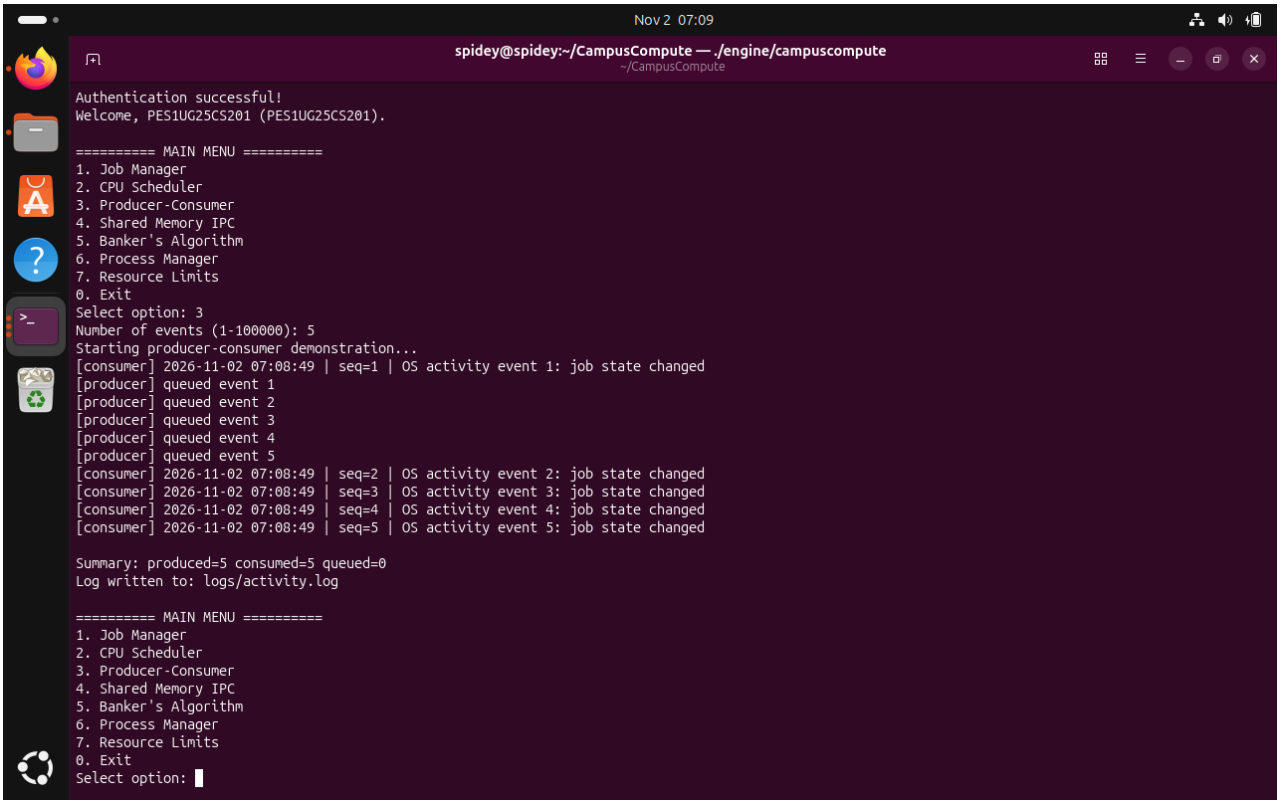
S21 – Part 2 of 4: Job 101 listed as READY, then marked RUNNING and listed again.



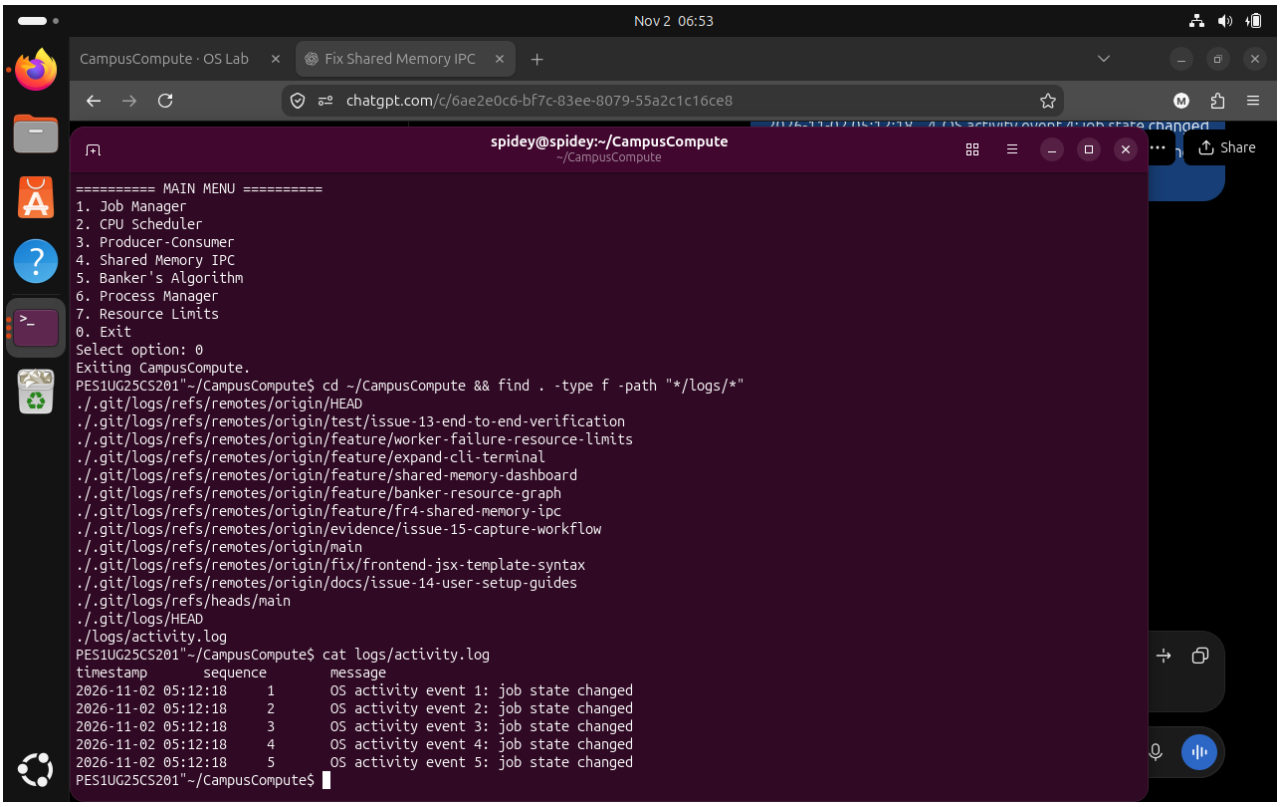
S21 – Part 3 of 4: Job 101 marked COMPLETED with completion time 12 and listed with status COMPLETED.



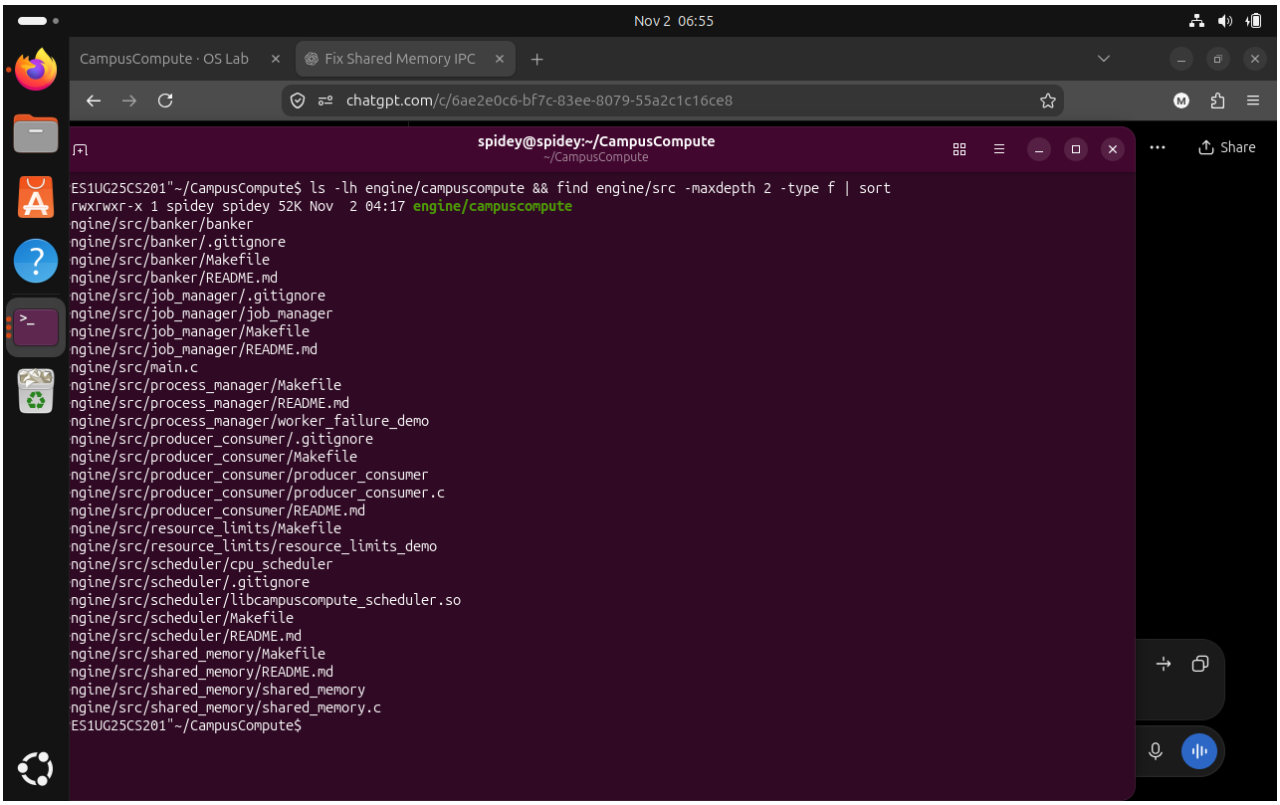
S21 – Part 4 of 4: Producer-consumer demonstration run within the integrated session: produced=5, consumed=5, queued=0.



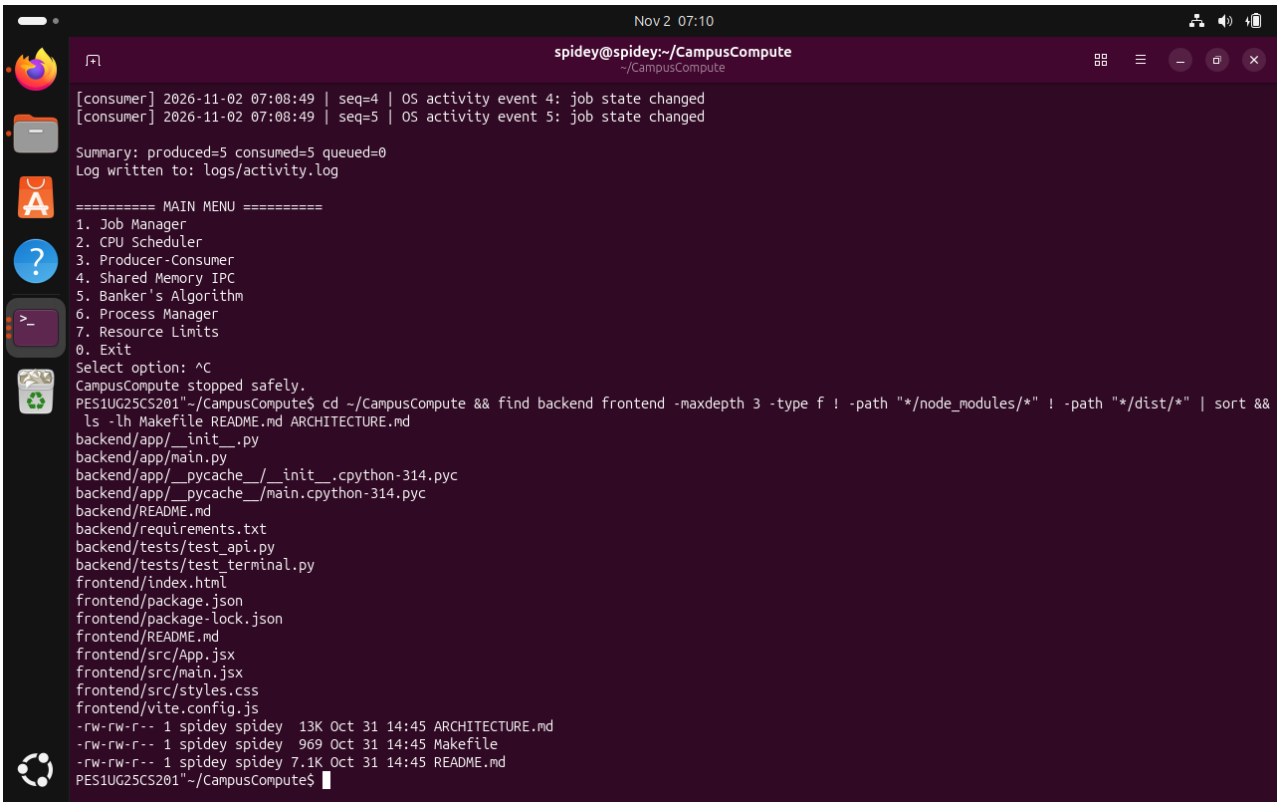
S22: logs/activity.log generated by the program, showing five timestamped and sequenced events.



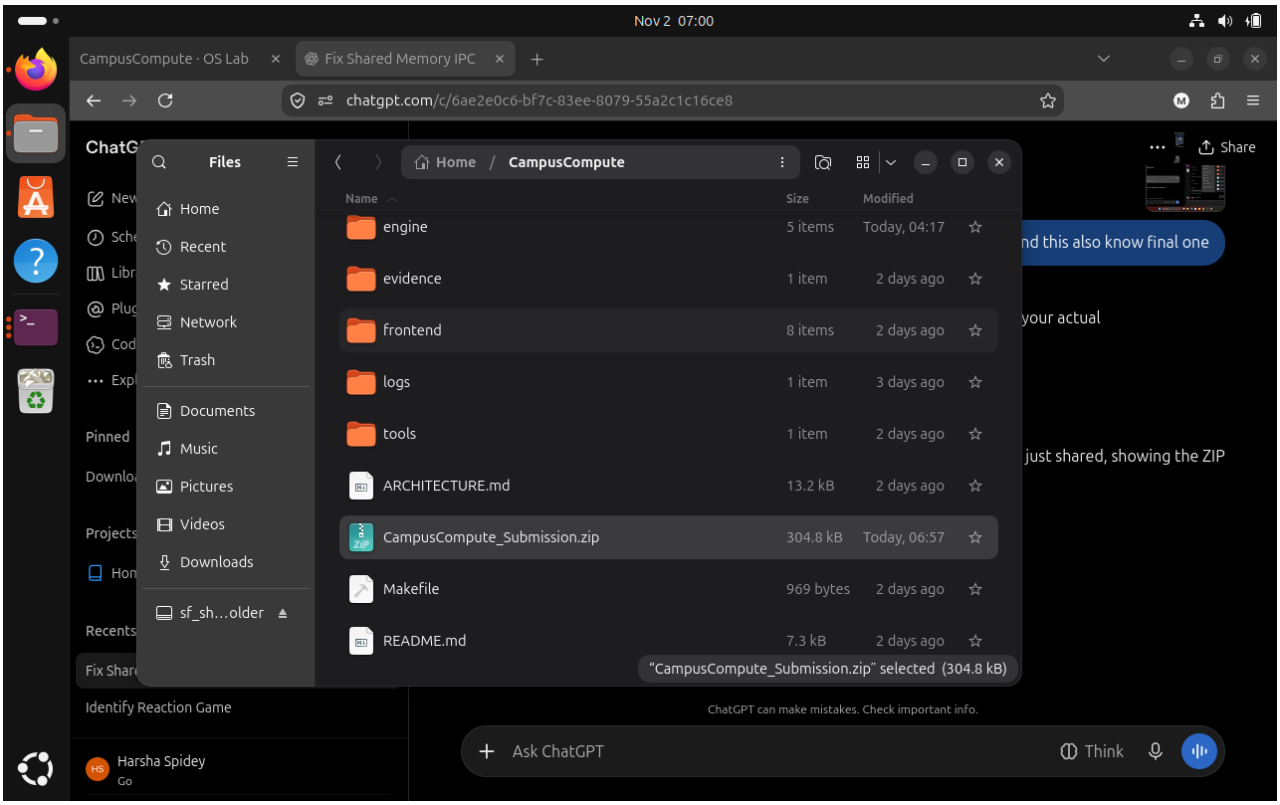
S23 – Part 1 of 2: Final engine source files and the compiled campuscompute executable.



S23 – Part 2 of 2: Final backend and frontend files together with the Makefile, README.md and ARCHITECTURE.md.



S24 – Part 1 of 2: The CampusCompute_Submission.zip archive in the project folder.



S24 – Part 2 of 2: Archive contents listed with unzip -l, and the integrity check reports no errors detected.

